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POLJOPRIVREDNI FAKULTET**



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VETERINARSKE MEDICINE SA MEĐUNARODNIM UČEŠĆEM**

**PROCEEDINGS
OF THE 43rd CONFERENCE FOR STUDENTS OF AGRICULTURE AND
VETERINARY MEDICINE
WITH INTERNATIONAL PARTICIPATION**

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**43rd CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY
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Council of the University of Novi Sad, Faculty of Agriculture, at its fourth meeting on 18th of February 1981 made a decision that the Conference for students of agriculture and veterinary medicine with international participation, held each year at the Faculty of Agriculture will be held in the memory of academician ***Dr Petar Drezgić***, professor, and the prizes awarded at the Conference will bear his name.

PREFACE

The first Conference for students of agriculture and veterinary medicine with international participation was held in Novi Sad 43 years ago. To the present date, many papers from national and international authors were presented (Hungary – Gödölö and Debrecen, Germany – Kassel, Slovakia – Nitra, Poland – Wroclaw and Warsaw, Romania – Timisoara, Cluj and Bucharest, Bulgaria – Plovdiv, Russia – Moscow and Kemerovo, Macedonia – Skopje and Bitola, Bosnia and Herzegovina – Sarajevo, Banja Luka, Croatia – Zagreb). Since 2009, papers presented at the Conference are published in a special publication named “PROCEEDINGS OF THE 43rd CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY MEDICINE WITH INTERNATIONAL PARTICIPATION” where all published papers have all elements of the original scientific paper. Starting this year, we enabled students and mentors who are interested in benefitting more from this Conference to publish their papers in the journal Contemporary Agriculture published by the University of Novi Sad, Faculty of Agriculture. Also, besides the Faculty of Agriculture from Novi Sad, co-organizer of the Conference is also Faculty of Agriculture from Timisoara and there is a prospect of joining another University center from the region. This Conference is strongly supported by the Student Parliament of the Faculty of Agriculture, Novi Sad. As the host and the principal organizer of the Conference University of Novi Sad, Faculty of Agriculture has a special honor and a pleasure to wish a warm welcome to all participants, successful work and amusing socializing in Novi Sad.

Vice-dean for science and international cooperation
Dr Branko Čupina, full professor

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A SURVEY ON RESPONSIBLE ANIMAL HUSBANDRY AMONG PET RABBIT HOLDERS

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Summary: *A survey was done with questioners. The query was focused on the following: how can become somebody an owner, what kind of breeds are the most popular, what kind of housing and feeding methods are used, what kind of animal health problems are occurred, how much is known of the biology of the rabbit, what is the fate of the rabbit. The aim of the study was whether pet keeping is appropriate for the domestic rabbit needs and welfare requirements. Most of the rabbits were kept in cages and were fed with pellet, but they are not as conscious in the prevention of diseases as in the donation of a pet.*

Key words: *pet, rabbit, animal care, welfare, survey*

INTRODUCTION

Millions of rabbits are kept as pet animals all around in Europe. People are moving to urban areas from the countryside and would like to preserve their contact with the nature and earlier life. Rabbit is one of the most popular pet (DeMello, 2010). The recent legal regulations of animal welfare are focus on the rabbit producers, and the regulation of protection of animals deal with rabbits in general. Nevertheless the conscious rabbit owners pay attention and money on their pets, e.g. insuring and microchipping their rabbits (Oxley et al., 2015). Nowadays, the objective relationship between humans and animals has ceased, and humans see animals as companions rather than objects (Jámbor, 2018).

Rabbits are prey animals, so they are more sensitive on their environment, sheds and disturbance. In other words, consideration should be given to environmental management to minimise disturbance as much as possible, but rabbits could be easily trained and conditioned to perform a series of behaviours (Bradbury and Dickens, 2016). So rabbits could be good companions if they are healthy and when stress-free, enriched living areas are provided them and welfare issues are daily applied (Rooney et al., 2014). At the same time several rabbits are kept in inappropriate conditions. A result of a survey in UK shows how many ways of housing are used among pet owners.

The wire mesh cages are more durable and easier to clean them than the wooden ones and they can be made more comfortable with many accessories such as resting shelf and chewing wooden block, feeders for pellet and hay (Szendrő et al., 2013). The rabbit needs a lot of water, so be sure to always have fresh water in front of it. It is best to fill the washable drink

bottles daily or even several times a day (Szendrő, 2016). The Animal Welfare Act states that outdoor animals should be given shelter when the weather is unfavourable and against predators. It even eliminates the idea of animal torture (Anonim, 2009).

Experience shows that solitary and in small places housed animals make more contacts with their owner than group-housed ones in a large system (Schepers et al., 2009). There are many cases when rabbits are raised together, but if we want to place two individually raised animals with one another, we need to use a step-by-step approach (Bourne, 2011).

The rabbit can suffer from many diseases (myxomatosis, RHD, Pasteurellosis, coccidiosis, ear mite, etc.). Try to prevent these and cure immediately in case of trouble (vaccination, hygiene, call a veterinarian) (Sinkovics, 2008; Vetési, 1990).

The animal owners should take into consideration the right of 5 freedoms of animals (Brambell, 1965):

1. „Freedom from hunger or thirst.
2. Freedom from discomfort.
3. Freedom from pain, injury or disease.
4. Freedom to express (most) normal behaviour.
5. Freedom from fear and distress.”

The aim of our study was to obtain information on the situation of pet rabbit keeping and respect of animal welfare recommendations in Hungary.

MATERIAL AND METHODS

Data were collected by paper and on-line questionnaires at the university and with the help of Facebook groups. The first group of questions were about the respondents' gender, age, location and type of living. The questions about their pet rabbit keeping were about: the way of get it, their opinion on gifting living animal, the number of animals, the way of housing of the rabbit, the feeding, the animal health problems, endurance, and sense of responsibility. Beside the closed and open questions a 5-point Likert-scale was used to collect data on the students' attitudes.

The data obtained from the questionnaire were evaluated by correlation and cross-table analysis using IBM SPSS 25 statistical software package.

RESULTS AND DISCUSSION

Responses (n=192) were received from almost all parts of Hungary, but mainly from Békés, Bács-Kiskun, Borsod-Abaúj-Zemplén and Jász-Nagykun-Szolnok counties. Due to the small sample the results are not representative. 79% of the total number of respondents has or had pet rabbit and 52% of them were female. The average age of them was 29.73 ± 13.96 years. The youngest was 10 and the eldest was 70 years old among the rabbit holders. 47-47% of the rabbit owners are living in villages and cities, only 6% of the questioners came back from Budapest. There was no difference between the rabbit holders whether they live in a family house or in an apartment. The respondents had primary (17.2%), secondary (60.3%) and higher education (22.5%).

The livestock holders kept one (28.48%), two (29.14%) or more (42.38%) rabbits at the same time. The rabbits were reproduced at 48% of those who had 2 or more rabbits ($r=0.651$, $p<0.01$).

The pet rabbits are kept in different type of cages in different places (flat 80% or shed 13%). Only 7% of them are kept in free range system.

Significantly more rabbits were fed with pellet (96.6%, $p<0.01$) than other feed types. The pelleted concentrate was usually supplemented with some grass or alfalfa hay. Where the rabbits were fed with other fodder cereals, hay, vegetables and apples were used. The most popular rabbit breeds are shown in Table 1.

Table 1: The rabbit breeds kept by the respondents

Breeds	Proportion (%)
Mixed breed	40.40
Dwarf rabbit breeds (not specified)	13.91
Lionhead rabbit	8.61
German giant	7.95
Hungarian giant	7.95
Castor rex	3.97
New Zealand White	3.31
Californian rabbit	3.31
Pannon white	2.65
Giant papillon rabbit	2.65
Flemish giant	1.99
Dutch rabbit	1.99
Hycole (hybrid)	1.32
Total	100.00

Source: Own data

The responsible animal keepers also took care of the animal's health. Rabbits have some dangerous infectious diseases for which a vaccine is available to prevent them. 55.22% of the owners asked for a vaccine against myxomatosis and some of them also against RHD

(Rabbit Haemorrhagic Disease). The most frequent diseases with which they went to the vet are in Table 2.

The perseverance and loyalty of keepers to the animal is strongly illustrated by the length of time they have been kept. Most of the respondents have been for years, but some have kept the animal for a few months or a few days only. On average, keepers keep rabbits for seven and a half years. The longest rabbit among the respondents has been a pet keeper for 41 years. The shortest time was 4 days for a pet rabbit.

We have come to the conclusion that men breed more frequent rabbits than women. (The correlation coefficient showed a weak relationship ($r = 0.220$).) Greater numbers of animals were donated by the people who not only kept but also bred them.

There is a weak relationship between the age at which people started to hold pet rabbit and their place of residence. In other words, those living in villages started to keep the animal sooner than those living in cities or in the capital ($r = 0.261$) and there is a weak relationship between where they live and the number of animals kept ($r = 0.220$).

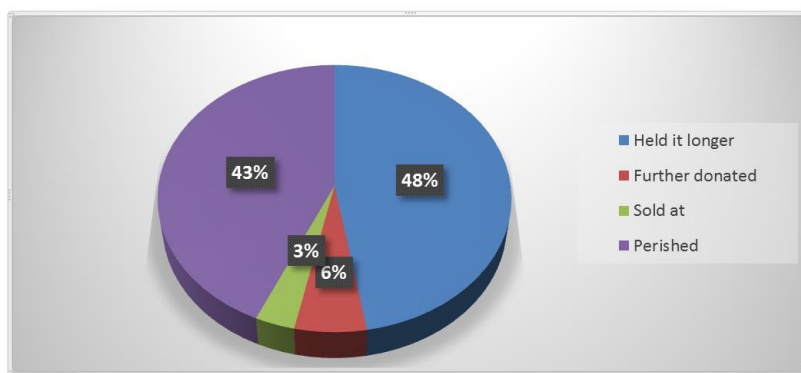
Table 2: The most frequent health problems

Breeds	Proportion (%)
Pasteurellosis	11.94
Enteritis	5.97
Ear mites	5.97
Coccidiosis	4.48
Diarrhoea	2.99
Conjunctivitis	2.99
Others (not identified)	4.48

Source: Own data

Among other things, it was asked how pet keepers are, or are willing to give rabbit as a present, and how aware they are of the responsibility of an animal. Surprisingly, only 29.8% of respondents would give or donate a pet to others, while the vast majority, 70.2%, would not want to. Here they had the opportunity to write an answer, which many have used. People who do not want to give a pet almost always gave the answer that an animal is a responsibility. Keepers think about the responsibility of giving a pet as a present to each other.

The fate of the rabbit changed in many ways (Fig. 1). There were livestock keepers who kept it longer, donated, sold or even the animal was dead after a while. Most respondents kept it longer. A small number is where the rabbits were given as gifts or sold.



Source: own data

Fig. 1: The fate of pet rabbits

CONCLUSION

Due to the low number of the sample the result of the survey is not representative, but it is a good preliminary result for planning further investigations.

Most of the people have the knowledge of the social behaviour and needs of the rabbits (Magnus, 2005). Only 28.48% of the respondents keep their animals on its own which result is better than Edgar and Mullan (2011) had.

The owners contact veterinary surgeon usually in case of symptoms of diseases similarly to other sources (Fisher, 2010; Mullan and Main, (2006), but slightly more than half of them care on prevention.

The general use of pellet is comfortable for the owner and fit to the nutritional requirements of the species.

There is a big debate on the use of cages in commercial rabbit production (Szendrő, 2016), but pet rabbits are usually kept in different type cages.

People typically receive animals as gifts for their birthday and Easter, but most of the respondents are thinking on a responsible way and would not like to donate living animals to those who are not ready for it.

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DETECTION OF SOIL-BORN FUNGI ASSOCIATED WITH WILTING OF POTATO AND PEPPER

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Summary: *Bosnia and Herzegovina has great potential and long tradition in the vegetable production. Samples from infected potato and pepper yield have been collected in the Gradiška Municipality area, Bosnia and Herzegovina. On pepper plants, we observed characteristic symptoms of wilting stopping on the increase, and a dark staining of the conduction tissue was observed at the cross section of the ground. In the field of the potato he observed: the wilt, the rot of the roots and ground part of the tree. The main goal of this research was isolating pathogens from infected potato and pepper samples and identification of those pathogens using classic and molecular methods (PCR). Based on researched pathogenic, morphological characteristics, it was established that acquired isolates belonged to Fusarium and Sclerotinia species.*

Key words: *Soil- pathogenic fungi, PCR, Pathogen isolation*

INTRODUCTION

Soil-born pathogens from genera Verticillium, Fusarium, Sclerotinia, Rhizoctonia and Pythium represents a serious problem in vegetable production as well as in open field and protected conditions. They could decrease total yield from 6,3 to 100 % (Katan, 2000; Bhat et al., 2003) what depends of hybrids other conditions. They are highly polyphagous and inadequate cultural and plant protection measures contribute their dispersal worldwide. There are several different measures used to prevent previously mentioned pathogens and chemical measures, and these measure are representing the complex integral protection program. However, the first step is the good diagnostic in order to know which which pathogens we are dealing. In 2019, several farmers reported wilting of potato and pepper plants in their production fields so the main aim was to check identity of the causal agents associated of wilting of pepper and potato studding pathogenic and morphological characteristics and perform molecular methods.

MATERIAL AND METHODS

Sampling and isolation

Potato displaying symptoms of wilting and stalk white mold and pepper xylem tissue wilting were observed during June 2019 in location Gradiska, Bosnia and Herzegovina, BiH. Isolation was performed from the collected samples using standard mycological methods (Dhingra and Sinclair, 1995). Fragments of diseased parts of the root and the ground part of the tree from the crossing of healthy and diseased the tissues were dissected with a sterile scalpel, washed under a jet of water, then surface-sterilized with 2% sodium hypochlorite (NaOCl) solution and transferred to sterile PDA (potato-dextrose agar) medium. Petri dishes were incubated in the dark, 26°C until the appearance of the mycelium. To study the microscopic characteristics, the isolate was seeded on SNA medium (Biley substrate, Jofe modified: 1 g KH₂PO₄; 1 g KNO₃; 0.5 g MgSO₄; 0.5 g KCl; 0.2 g glucose; 0.2 g sucrose; 0.2 g starch; 15 g agar; 1 l distilled water) (Levic, 2008). Isolated were identified studding macroscopic and microscopic morphological characteristic isolates after 7 and 15 days after isolation.

Pathogenicity test

Pathogenicity tests were carried out on disease free pepper and tomato seedlings (4-5 leaves) by watering in the root zone with the inoculum of the observed mycelia developed on PDA and SNA medium. The inoculum was made as a suspension by adding 10 ml of distilled water to a Petri dishes where the 5 days old mycelium was previously added. The inoculated plants were covered with plastic bags and after 3 days of incubation, in order to fulfil the Koch's postulate, the pathogen was re-isolated from symptomatic plant tissue.

PCR

The DNA extraction was conducted from mycelium sampled directly from colonies growth on PDA medium using Qiagen PlantDNeasy mini kit (Qiagen, Germany) following the procedure recommended by the manufacturer. PCR reaction was performed in total volume of 25 μ l using primer pairs ITS1 (5' TCC GTA GGT GAA CCT GCG G 3') and ITS4 (5' TCC TCC GCT TAT TGA TAT GC 3') (White et al., 1990) which amplifying five genomic DNA fragments from ITS region rDNA (fragments of 18S rRNA, ITS1, 5.8S rRNA, ITS2 and part of 28S rRNA). PCR products were visualized on 1 % agarose gel in 1xTBE buffer (90 mM Tris; 90 mM Boric acid; 1 mM Na₂EDTA) at 100 V. Selected PCR amplicons were sequenced in Macrogen Inc. (Europe). Obtained sequences were analyzed and assembled BioEdit software (Hall, 2011) and compared in BLAST (Basic Local Alignment Search Tool) (<http://www.ncbi.nlm.nih.gov>) with available sequences in NCBI GenBank.

RESULTS AND DISSCUSION

On diseased potato plants collected from the Gradiska location, and delivered pepper samples, symptoms were manifested in the form of plant fading, rotting of the root and ground part of the tree, chlorosis, necrosis, premature leaf loss, dwarfism. In plants with symptoms of "wilting caused by *Fusarium*", on the cross-section basal stem was observed eclipse coloration in vascular tissue, and in some diseased plants in conditions of high humidity developed a white, woolly mycelium on the ground floor of the tree. (Figure 1, Figure 2). The isolates were identified based on the studied morphological characteristics of the colony grown on PDA medium. Morphological characteristics were observed: mycelium type, appearance and color of the colony, characteristics of the colony edge, presence of different scattering structures. Isolation of pathogens from pepper samples, growing on a PDA substrate, forms a white, air-uniform cotton mycelium that is observed at 26 °C, seven days after seeding. Three to five days after seeding, pinkish-purple pigment in the nutrient medium were observed (Figure 3). On SNA medium, in the 5-day-old culture showed slightly sickle elongated and narrow macroconidia as well as single-celled elongated microconidia (Figure 4). Based on the studied characteristics, it was determined that the test isolate belongs to the species from genus *Fusarium*. From potato isolates two forms of mycelia growth were observed on PDA medium at 25 °C, pinkish-purple where macro and micro conidia of *Fusarium* spp. was observed and a well-developed white mycelium that is initially homogeneous and airy, whereas later observed cottony radially spaced clusters of hyphae (Figure 5). Four days after seeding, macroscopic observation revealed the presence of black structures/sclerotia (Figure 6) pointed out presence of *Sclerotinia sclerotiorum*.



Figure 1. White mycelium the potato stalk



Figure 2. Eclipse coloration on pepper

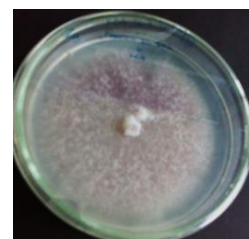


Figure 3. Pinkish-purple on mycelium

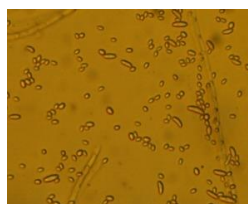


Figure 4. Macroconidia and Microconidia (60x)

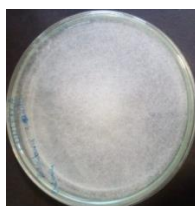


Figure 5. Cottony radially hyphae of *Sclerotinia*



Figure 6. Black sclerotia

Pathogenicity test

From diseased pepper plants, a greater number of isolates were obtained, of which three were selected for further testing based on morphological characteristics. On inoculated tomato and pepper seedlings, after incubation for 7-15 days at 20-25 C, fading, root rot and wilting were observed (Figure 7, Figure 8). To fulfil the Koch's postulates, pathogen was re-isolated from symptomatic plant tissues and the structure specific for *Fusarium* spp and *S. sclerotiorum* were observed.



Figure 7. Inoculated pepper seedlings after 10 days



Figure 8. Inoculated tomato seedlings after 10 days

PCR

Three isolates were submitted to the PCR reaction with ITS1/ITS4 primers (White et al., 1990). Obtained PCR amplicons results in nucleic acid fragments sizes around 600-650 bp in potato and pepper isolates indicating presence of *Fusarium* spp. and 550 bp in DNA from potato isolate showing presence *S. sclerotiorum*. Comparison of obtained sequences with other sequences from GenBank using the BLAST tool confirmed the identity of the isolate from pepper 100% with *Fusarium oxysporum* isolates from sweet potato (KU382557 and KU382558) from North Caroline, USA while isolate from potato showed highest identity with *Fusarium graminearum* isolate from *Eupatorium chinense* L, China (MK243483). Sequence from isolate of *S. sclerotiorum* didn't analyzed in the study.

Pathogens from soil represent a significant problem in the cultivation of vegetable plants both in the protected area and in the open field. Koike et al. (2003) found that species of the genus *Fusarium*, *Sclerotinia*, *Rhizoctonia* represent one of the most common soil pathogens that threaten plant production. Research conducted in Serbia confirms these allegations, where it has been established that pathogens of the genus *Fusarium*, *Sclerotinia* cause damage in the production of peppers in Serbia (Mihajlović et al., 2017). Several *Fusarium* species viz., *F. oxysporum* and *F. solani* have been reported as major pathogens causing wilt disease in solanaceous crops like chilli, tomato, brinjal in Gujrat and Uttar Pradesh in India (Naseem et al., 2014). The wilt/root rot incidence in chili growing areas varied from 10 to 54% (Dashora and Sharma, 2018). There are certain similarities between symptoms caused by *F.oxysporum* and symptoms caused by other fungi, and because of those similarities, it is necessary to conduct microscopic determination in order to properly determine the diagnosis (Ivanović, 2004). *Sclerotinia sclerotiorum* (Lib) De Bari is a polyphage phytopathogenic

fungus. It belongs to the *Ascomycota* division, the classic *Discomyceta*, the *Helotiales* order, the family *Sclerotiniaceae*, the genus *Sclerotinia*. The cause was plant rot, which occurs in temperate areas (Saharan and Mehta, 2008). The control of diseases caused by *S. sclerotiorum* depends mainly on chemical fungicides, since no resistant cultivars are available. Therefore, it is important to explore novel, environmentally friendly alternatives to chemical fungicides or to reduce amounts of chemical fungicides used (Hamid, et al., 2018). The soil-borne fungus *S. sclerotiorum* infects many important crop plants. Central to the success of this pathogen is the production of sclerotia, which enables survival in soil and constitutes the primary inoculum. (Taylor et al., 2018). *Fusarium graminearum* is best known as a pathogen causing head blight (scab) on wheat and barley, and earrot in corn, but in recent years, has been implicated as the cause of other plant diseases in North America, sugarbeet yellows (Hanson, 2006), soyabean root and seedling rot (Broders et al., 2007) in the USA, and pea root rot in Canada (Feng et al., 2010). *F. graminearum* were reported to cause dry rot when inoculated on potato tubers in Tunisia (Daami-Remadi et al., 2006) and Canada (Peters et al., 2008). These observations of additional hosts of *F. graminearum* have epidemiological implications since cereals, potatoes, sugarbeet and soyabeans, for example, are frequently grown (Estrada et al., 2010).

CONCLUSION

Pathogens isolated from symptomatic pepper and potato plants, examining their pathogenic morphological (microscopic and macroscopic) and molecular characteristics, determined that they belong to the species: *F. oxysporum* (pepper sample), *S. sclerotiorum* and *F. graminearum* (potato samples). The symptoms of soilborne disease, caused by different pathogens, are very similar. They include root rot, root blackening, wilt, yellowing, stunting or seedling damping-off, bark. Soil-borne species can survive for many years in the absence of a host plant by forming resistant structures such as: microsclerotia, sclerotia, chlamydospore. Consequently, these pathogens are particularly difficult to predict, detect, diagnose and successfully control.

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WELFARE PARAMETERS AND KEEL BONE DAMAGE IN LAYING HENS REARED IN DIFFERENT PRODUCTION SYSTEMS

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Summary: *The aim of this study was to determine the effect of housing system on welfare parameters and the prevalence of keel bone damage in laying hens. In this study two housing systems were evaluated: aviary system and enriched cages. Results of this research showed that housing system has significant effect on prevalence of keel fractures ($P=0.0078$) and footpad dermatitis ($P=0.024$) in laying hens. Footpad dermatitis had statistically higher occurrence in enriched cages. It can be concluded that housing system have significant effect on prevalence of keel fractures and welfare parameters.*

Key words: *keel bone damage, housing system, welfare, laying hens, aviary, enriched cages*

INTRODUCTION

Animal welfare is a very important issue which can be defined as animal's body and mind state, and the extent to which its nature is satisfied (*Duncan and Fraser, 1997*). In general there are 3 aspects of animal welfare. First one is animal health, second one is production, and the third one is how an animal is feeling. Based on those facts one cannot conclude that if animal is healthy and producing well its welfare is not endangered. On the contrary, it is known that animals have feelings, and all three aspects needs to be satisfied, which presents ethical and practical problem.

In 2012 EU countries banned conventional cages for the welfare reasons, when Directive 1999/74/EC came to force. In Serbia this directive will come to force in 2020. This will lead to transitioning to new housing systems in poultry production which are significantly better when it comes to animal welfare. However, new systems generated new problems, such as keel bone damage (KBD), which also endangers hen's welfare. Keel bone damage can be described as deviations (KD) and fractures (KF), and both can be painful for birds and reduce productivity (*Harlander et al., 2015*). Fractures are characterized by sharp bends, shearing and fragmentation of the keel bone. Deviations are characterized by an abnormally shaped structure that varies from a theoretically perfect 2-dimensional straight plane (*Casey-Trott et al., 2015*). It ranges from 5% to as high as 97% depending on housing system and age of hens (*Rodenburg et al., 2008; Wilkins et al., 2011; Petrik et al., 2015; Riber and Hinrichsen, 2016; Regmi et al., 2016*). Even old fractures, resulting in formation of callus, can lead to

chronic pain, which can reduce productivity and welfare (*Sandilands et al., 2009; Lay et al., 2011; Hardin et al., 2019*).

Besides the keel bone condition, there are other welfare parameters which can indicate the well-being of the birds. *Nicol et al. (2009)* described plumage score as the most important welfare indicator in laying hens because feather pecking is one of the most important causes of feather damage, and presents major welfare problem (*Habig and Distil, 2013*). Footpad dermatitis and skin lesions are a form of contact dermatitis, affecting skin in contact with irritating materials (*Greene et al., 1985*). The footpad condition is an important aspect of poultry welfare that in severe cases can cause pain (*Berg, 2004*).

The aim of this study was to determine the effect of housing system on welfare parameters and the prevalence of keel bone damage in laying hens.

MATERIAL AND METHODS

In this research two types of alternative housing systems existing in Serbia were examined: enriched cages with full equipment and aviary system. In this research 50 hens were randomly selected hens from each housing system. Hens were at the end of the production cycle (older than 60 weeks of age). In cage system hens were taken from different cages and levels, and from aviary system hens were taken from the floor and different tiers.

Assessment of keel bone damage was done by palpation. The prevalence of KBD was assessed by using the technique of palpation according to the method described by *Wilkins et al. (2004)*. Palpation was done by running fingers alongside and over the keel bone. It was determined whether the damage was present or not.

Welfare parameters included in this research were: plumage condition, occurrence of footpad dermatitis (FPD), skin lesions, comb pecking wounds, bumble foot and claw length. Welfare parameters were assessed by visual method. Plumage condition was assessed by using the method described by *Tauson et al. (2005)*, scale ranging from 0 (highly damaged plumage) to 4 (very good plumage). Footpad dermatitis, skin lesion and comb pecking wounds were assessed on the scale from 0 to 2 depending on severity (0 presenting no visible damages, and 2 presenting severe damage). Claws were assessed by its length, and were described either as normal or long.

Statistical analyses were performed using Statistica 13.5 using t-test. Results of statistical analysis were considered significant when the $P < 0.05$.

RESULTS AND DISCUSSION

The results of this research showed high prevalence of KBD in both housing systems, as well as high prevalence of footpad dermatitis in enriched cages.

The higher prevalence of KBD was detected in aviary system ($P < 0.05$) compared to enriched cages. Besides that, the main difference in these two housing systems is the type of KBD. In aviary system main cause of KBD were keel fractures (KF), and in enriched cages

the main type of KBD were keel deviations (KD). In aviary systems hens hitting the equipment more than in other systems (Riber *et al.*, 2018). This might be the explanation for the higher percentage of birds with fractures in aviary system compared to enriched cages. Đukić Stojčić *et al.* (2017) reported that 3% of birds in enriched cages without equipment were affected by KBD, while 39% of hens were affected in fully equipped cages. They concluded that metal perches are the main reason of the increased prevalence of KBD. The assumption that perches have a key role in developing KBD was confirmed by other authors too (Rodenburg *et al.*, 2008; Wilkins *et al.*, 2011). Wilkins *et al.* (2011) found an increase in KBD between 10 and 37% in mobile organic houses when perches were added.

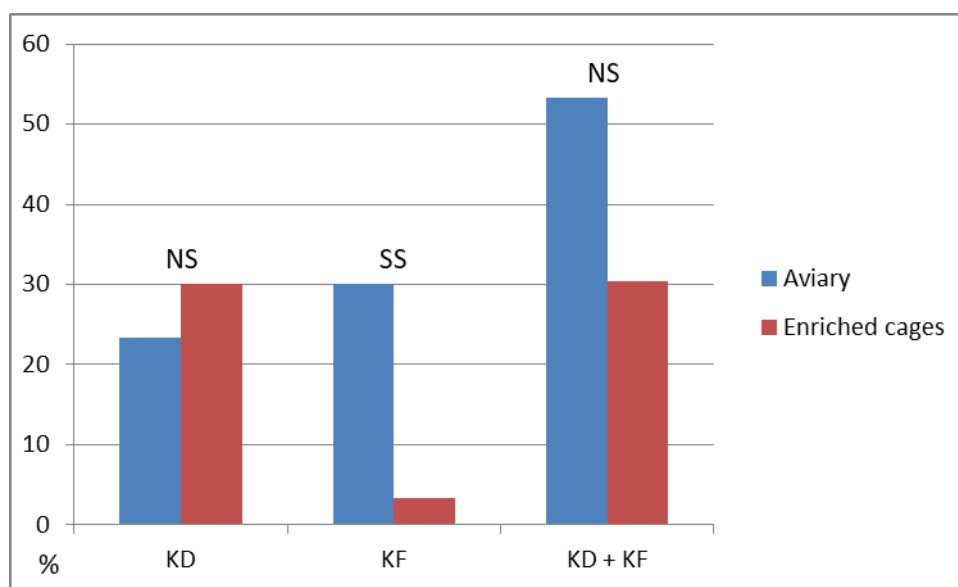


Fig. 1. Prevalence of KBD in laying hens in aviary system and enriched cages. KD – keel deformation, KF – keel fracture, NS – statistically nonsignificant, SS- statistically significant ($P < 0.05$).

Higher prevalence of footpad dermatitis was found in enriched cages. Statistical analysis revealed statistically significantly higher prevalence of FPD in enriched cages ($P = 0.024$). Over 33% of the birds in enriched cages had FPD which is in conflict with the findings of other authors. For example, Neibuhr *et al.* (2009) found slightly higher prevalence of FDP in non-cage systems. Rönchen *et al.* (2007) found that between 25 and 37% of birds, in 3 different flocks, had footpad lesions reared in aviary system.

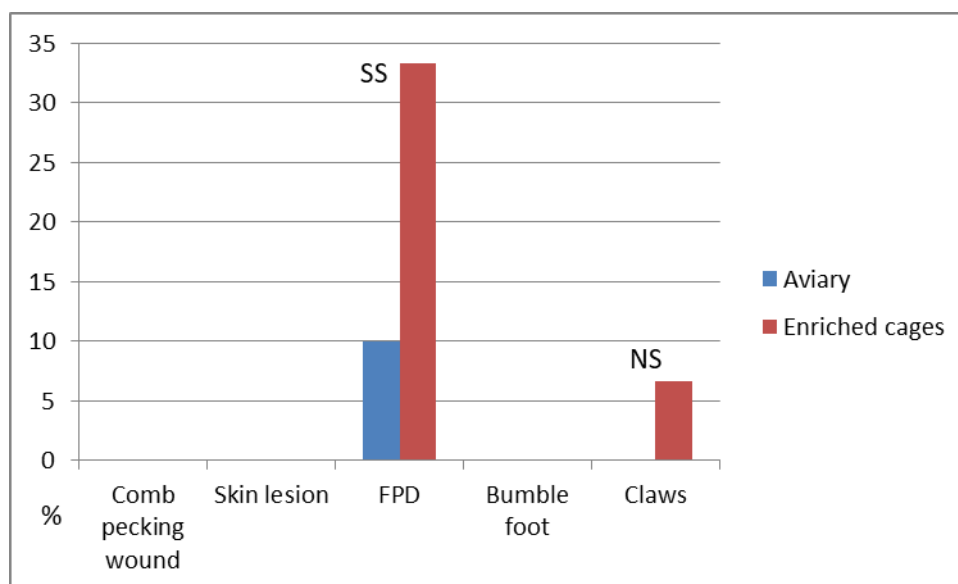


Fig. 2. Welfare parameters in laying hens in aviary system and enriched cages. FPD-foot pad dermatitis, NS - statistically nonsignificant, SS - statistically significant ($P < 0.05$).

No hens were found with comb pecking wounds, skin lesions or bumble foot in both housing systems. As for the claws length, birds with the long claws were detected only in enriched cages (7%). Normal claw length can be explained by existence of claw shortening devices.

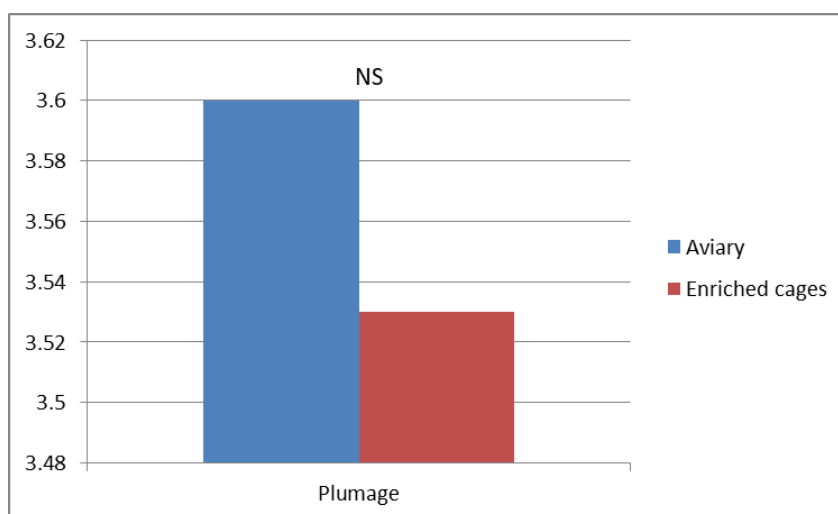


Fig 3. Mean score of plumage condition in aviary system and enriched cages. NS - statistically nonsignificant, SS - statistically significant ($P < 0.05$).

Plumage condition, as one of the most important welfare parameter, was not significantly different between two housing systems. However, slightly better plumage condition was detected in laying hens housed in aviary system. *Rönchen et al. (2007)* reported that housing system and its furnished elements can cause feather damages in laying hens.

CONCLUSION

Significant differences in occurrence and the type of KBD were found between the housing systems. The higher occurrence of KBD was found in aviary system compared to enriched cages. Regarding the type of the KBD higher prevalence of keel fractures was found in aviary system, while more keel deviation was found in enriched cages. In addition, higher incidence of footpad dermatitis was found in enriched cages.

Further research is needed to determine specific risk factors of KF, KD and its association with KBD in order to develop strategies for reducing the incidence of this multifactorial welfare issue.

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ESTIMATING THE ECONOMIC LOSSES IN DAIRY PRODUCTION, BASED ON MILK QUALITY

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Summary: *One of the challenges farmers face is how to compete in the market by producing quality milk, but also to reduce milk production losses. The purpose of our research was to determine the economic losses caused by changes that occur on milk quality (protein and fat) and hygiene (somatic cell count (SCC/ml) and total number of bacteria (CFU/ml) in dairy milk production. The results obtained from the analyzed samples from the four farms in the Pelagonia region indicate that these parameters as quality measures vary from producer to producer and are a result of insufficiently implemented milk quality control.*

Key words: *raw cow's milk, losses, somatic cells.*

INTRODUCTION

The chemical composition of raw milk (Guetouache M. et al., 2014), especially the percentage of milk fat and protein, are important parameters for its quality. Consequently, any decrease in their nutrition or poor health would result in lower quality milk production. In order to produce milk that will be a quality raw material, in addition to its chemical composition, special attention must be paid to the health of the cows and the udder. The total number of bacteria (CFU/ml) and the total number of somatic cells (SCC/ml) are direct indicators that the cow is infected with mastitis (Forback L. et al., 2010).

The simplest and easiest way to detect clinical mastitis is physical control of the udder (FAO, 2014). The diseased udder is swollen, palpable and reddened, and the secreted milk is most often mild or has clumps of milk (Makarijoski B. et al., 2015). On farms where Good Hygiene and Manufacturing Practice is well conducted (Trajkovska B. et al., 2016) analysis of the total number of bacteria in raw milk is a good way to detect the possible occurrence of mastitis. The most common causes of this type of infection are bacteria such as: *Streptococcus agalactiae* and *dysgalactiae*, *Staphylococcus aureus*, and in addition *E. coli*, *Enterobacter* and *Klebsiella* from coliforms (Asfaw M. and Negash A., 2017). The increased number of somatic cells in the milk is a reliable indicator of the inflammatory process of the udder (Musliu A. et al., 2009). Approximately 98% of somatic cells in milk are leukocytes, which increase as a result of bacterial invasion, and the remaining 2% are epithelial cells resulting from umbilical regeneration (Kochoski Lj., 2011).

MATERIALS AND METHODS

The research was conducted over a period of three months in four farms in Pelagonia region. Samples for analysis of the bulk tank raw cow's milk were taken once a month, in accordance with official dairies control for monitoring the quality of raw milk. A total of 24 samples were analyzed. Samples for the analysis of protein and milk fat percentages were taken in plastic cups. Reference methods were used for both analyzes, the protein percentage was analyzed by the Kjeldahl method (MK EN ISO 8968-1: 2001) and the fat by the Gerber ISO 488: 2008 method (IDF 105: 2008). Samples for analysis of milk hygiene were taken in sterile bottles with Broad Spectrum Microtabs II preservative (one tablet for 40 ml of milk). The total number of bacterial count (CFU/ml) was determined by the colony count method MK EN ISO 4833: 2003, while the total number of somatic cell count (SCC/ml) was determined by LACTOSCAN SCC apparatus. After sampling the samples were well sealed and transported at 4 °C to the laboratory, in accordance with the *Rulebook on how to store raw milk and to take samples for analysis and superanalysis, the content, form and manner of filling in the minutes, the methods of analysis of raw milk, the method of classification and method of evaluation of raw milk, as well as the method of training and the program for conducting the training for controllers* (Official Gazette of R.M.No. 151/2011). The examinations were completed during the same day.

The determination of the price per liter of raw milk that should be obtained by the producer was carried out in accordance with the formula published in the marketing authorization (Official Gazette of the Republic of Macedonia No. 151/2011):

$$PPM = (M \times mp) + (P \times pp) + K,$$

where, PPM - purchase price of raw cow's milk; M -% of milk fat; P -% of protein; mc - price for milk fat percentage; pp - price for percentage of protein; K - Correction (quality supplement).

In the formation of the purchase price of raw milk (PPM) milk fat should participate with 40%, and milk proteins with 60%. In addition, a quality adjustment of PPM will be made, where the correction (%) will be + 10% for milk with extra quality, 0% for Class I and -10% for Class II milk. The calculations that will be done in this paper will refer to the average purchase price of 17 MKD per liter of raw cow's milk, for Class I milk, with 3,5% milk fat and 3,2% milk protein, where as extra-class milk - milk containing up to 100.000 CFU/ml and up to 400.000 SCC/ml is taken. Milk with a total CFU/ml of between 100.000 and 200.000 and a SCC number not exceeding 400.000/ml is classified as Class I milk, while milk exceeding these limits is classified as Class II.

RESULTS AND DISCUSSION

There are a number of published papers worldwide about the economic consequences of changes in the physico-chemical composition of milk and hygiene propriety. From an economic point of view, the purchase price of raw cow's milk is the sum of the losses from reduced milk yield or poor quality milk and the costs of veterinary surveillance and animal treatment (FAO, 2014). According to Hogeveen H. (2011) the increase in milk producers' losses is a result of minimal investment and inadequately implemented Good Hygiene Practice, while expenditures are inevitable. According to Petkovski B. (2019) surveys, only 4,01% of farms meet the conditions for extra-class milk production, i.e. in accordance with the *Rulebook on the special requirements for safety and hygiene and the manner and procedure of official controls on milk and dairy products* (Official Gazette of RM 26/12, 145/14, 59/16, 197/16 and 16/2019) category Ist farms, or are eligible for extra quality milk production.

Table No. 1 - Spreadsheet overview of the results of the quarterly control of raw cow's milk at the first farm

Farm No. 1	First control	Second control	Third control
% of milk fat	3,52%	4,07%	4,12%
% of protein	3,43%	3,60%	3,39%
CFU/ml	304.879	426.996	1.304.581
SCC/ml	385.987	374.094	419.531
Milk yield (L) on a monthly basis	5.433	5.360	4.446
Purchase price of raw milk (MKD)	19,37	19,56	17,31
Total monthly price (MKD)	56.831,58	49.897,56	43.413,48
Monthly change compared to first control (MKD)	/	- 6.934,02	- 13.418,1

Satisfactory chemical composition and hygienic quality of raw milk at the first and second controls result in increased milk yield and milk purchase price (farm 1). Whereas in the third control the increased total number of bacteria and number of somatic cell counts are thought to be the result of mastitis infection (Forsback L. et al., 2010) and they are the cause of the reduced milk yield as well as the reduced milk purchase price (farm 1).

Table No. 2 - Spreadsheet overview of the results of the quarterly control of raw cow's milk at the second farm

Farm No. 2	First control	Second control	Third control
% of milk fat	3,70%	4,24%	4,29%
% of protein	3,43%	3,46%	3,58%
CFU/ml	652.610	691.010	330.655
SCC ml	232.161	285.736	215.169
Milk yield(L) on a monthly basis	2.508	2.551	2.934
Purchase price of raw milk (MKD)	18,13	19,74	20,15
Total monthly price (MKD)	45.470	50.356	59.120
Monthly change compared to first control (MKD)	/	4.886	13.650

Increasing the total number of bacteria and somatic cells, in addition to reducing milk production (Makarijoski B. et al., 2015), reduces the purchase price of milk. Starting from the fact that the farm's low hygiene also affects the health of the milking heads (Petkovski B., 2019), the hygiene improvement was expected and the relatively higher purchase price of milk in the third control (farm 2).

Table No. 3 - Spreadsheet overview of the results of the quarterly control of raw cow's milk at the third farm

Farm No. 3	First control	Second control	Third control
% of milk fat	3,58%	3,54%	3,50%
% of protein	3,35%	3,15%	2,96%
CFU/ ml	295.566	230.751	85.585
SCC/ml	235.724	320.393	291.395
Milk yield(L) on a monthly basis	6.325	6.632	5.168
Purchase price of raw milk (MKD)	18	17,05	16,52
Total monthly price (MKD)	113.850	113.075	85.375
Monthly change compared to first control (MKD)	/	- 775	- 28.475

Table No. 4 - Spreadsheet overview of the results of the quarterly control of raw cow's milk at the fourth farm

Farm No. 4	First control	Second control	Third control
% of milk fat	4,13%	4,24%	4,26%
% of protein	3,24%	3,25%	3,31%
CFU ml, for period	334.118	99.897	110.607
SCC ml, for period	410.730	454.075	334.234
Milk yield(L) on a monthly basis	3.739	4.524	4.011
Purchase price of raw milk (MKD)	17,05	17,85	19,47
Total monthly price (MKD)	63.750	80.753	78.094
Monthly change compared to first control (MKD)	/	17.003	14.344

The reduction in the milk purchase price at the third and fourth investigated farms is a result of the decrease in the percentage of milk fat and protein which are important parameters when forming it in accordance with the marketing authorization (Official Gazette of RM no. 151 / 2011). In addition, the hygienic integrity of the milk is satisfactory.

Corresponding to the classification of raw milk tested is the Musliu A., (2009) study, where in order to calculate financial losses, a correction factor based on the requirements of the Ministry of Agriculture, Forestry and Rural Development of Kosovo was used (MAFRD Administrative Instruction MA 20/2006). Loss in third-class milk producers results in a reduction of the basic price by 20% (the purchase price is 80% of the normal price), while the purchase of the extra class milk increases the basic price by 15% (the purchase price is 115% from the standard price). On the other hand, milk producers whose milk contains up to 400.000 SCC/ml have a loss in milk production of 1,4L (8%), and those who produce milk with about 800.000 SCC/ml incur losses of 2L (11%) per head (Kalevska, T., 2009). The high total number of somatic cells i.e. reduced milk production requires additional veterinary services that increase costs in order to reduce losses (Asfaw M. and Negash A., 2017).

CONCLUSION

For the successful operation of the milk producers and the economic viability it is essential to control the chemical composition and hygienic quality of the milk. Milk quality changes results in a considerable reduction in a profit in Macedonian farms. On the other hand, increased number of somatic cell count and total bacteria count may reflect on a presence of mastitis on dairy farms and this result more on less persistent decrease in milk yield.

In practice, the purchase price of raw cow's milk is not established in this way, i.e. the purchase price is the result of a previously concluded agreement between the producer and the dairy industry. If the requirements of the Rulebook (Official Gazette of the Republic of Macedonia No. 151/2011) were applied, changes in the milk purchase price on a monthly

basis would force farmers to change their approach to the raw milk they produce, and would address much greater attention to improving hygienic conditions on farms. This is an important step in motivating farmers to implement effective management practices aimed at improving milk quality and reducing the losses.

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COMPETITIVENESS OF EXPORTS OF THE SOYBEAN INDUSTRY FROM THE REPUBLIC OF SERBIA

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Summary: *The subject of the research is competitiveness of exports of the soybean industry in the Republic of Serbia in the period between 2009 and 2018. By using the indicators of competitiveness (RCA, RCA1 and RCA2), it can be concluded that the export of soybean in the Republic of Serbia has got a certain number of competitive advantages. Despite the fact that soybean and its products have positive influence on the foreign trade balance of the Republic of Serbia, there are measures to improve the sector. Implementation of these measures would have major impact on the economy.*

Key words: *competitiveness, soybean, export*

INTRODUCTION

In the last 60 years in Europe, as well as in the world, there has been exponential growth in the demand for plant-based protein sources. This growth is mostly related to the increase in consumption of meat and dairy products. Soybean is very important in livestock production as it is used in animal feeds due to its high content of protein. In Europe, over 70% of the needs for food with high protein value are settled with soybean and its product – soybean meal (Đorđević et al., 2015). However, on the global market European countries produce only between 1% to 2% of this legume (Malidža et al., 2016). Globally, the biggest producers are: the USA and the countries of the South America, among which Brazil is the leader. Although livestock production is highly developed in European Union, the production level of soybean is low. It is easy to conclude that the import of soybean in the European Union is high, due to the deficit of the plant-based protein in the European Union. This deficit has a negative impact on the autonomy of agriculture and the food industry (Murphy, 2018). On the other hand, the Republic of Serbia is a leader in production of non-genetically modified soybean in Europe. As far as genetically modified (GMO) soybean is concerned, the Republic of Serbia is on the forth place on the market. This research will focus on the competitiveness of the Republic of Serbia in the export of soybean on the international market. The aim of the research is to determine the position which the Republic of Serbia takes on this constantly growing market in Europe, as well as the options for further development on the market.

MATERIAL AND METHODS

Methods used in this research include the methods typically used in economics: analytical methods that are based on the analysis of the previous experience. Data processing is based on the statistical methods such as time series analysis which is performed using a graphical method. The data used in this research include publications by the Statistical Office of the Republic of Serbia, Food and Agriculture Organization of the UN (FAO), UN Comtrade Database, World Bank and International Trade Statistics. Other data sources used in this paper are listed in References. The research is based on the period between 2009 and 2018. In the analysis of competitiveness of the export of the soybean industry in Serbia, several important indexes are used. The indexes that are often used in this particular field of economy and in this paper are: Revealed Comparative Advantage Index or Balassa Index (RCA), Competitiveness Growth Index (RCA1) and Index of Net Business Performance (RCA2). RCA index can be modified to determine the competitiveness of the observed product in the sector. For this paper, the intersectoral competitiveness of the soybean will be calculated. Revealed Comparative Advantage Index (RCA) is calculated by the following formula (Balassa, 1965):

$$RCA = \log ((X_{ij}/M_{ij})/(X_{wj}/M_{wj})),$$

where:

X_{ij} is the export of j product (sector) of the country i ; M_{ij} is the import of j product (sector) of the country i ; X_{wj} is global exports of the product (sector) j , and M_{wj} is global imports of the product (sector) j . If the RCA index is less than 0, the observed product has a lack of competitive advantages. When the value of the index is between 0 and 1, there are competitive advantages. When the value of RCA is more than 1, the observed product or the industry is competitive.

RCA1 index is based on the RCA indicator, but it can also show competitiveness on the national level (Aigner et al., 2012) and it is calculated by the following formula:

$$RCA1 = \log ((X_{ij}/X_i)/(X_{wj}/X_w)),$$

where:

X_{ij} is the export of j product (sector) of the country i ; X_i is total exports of the country i ; X_{wj} is global exports of j product (sector), and X_w is total global exports. If the value of RCA1 is more than 1, there is competitive advantage of the observed product on the global market. But if the value of the index is less than 1, the observed product has a lack of competitive advantage on the observed level.

RCA2 index shows the contribution of an industry in forming a payment balance. In other words, the index shows the competitive advantage of export of an industry or a particular product (Balassa, 1965) and it is calculated by the following formula:

$$RCA2 = (X_{ij} - M_{ij}) / (X_{ij} + M_{ij}),$$

where:

X_{ij} is the export of j product (sector) of the country i , and M_{ij} is the import of j product (sector) of the country i . If the value of RCA2 equals -1, the export of the observed product

does not exist. If the value of the index is between -1 and 0, the observed product has a lack of competitive advantage. If the value of RCA2 equals 0, the export is equal to the import. If the value of this index is between 0 and 1, there are competitive advantages. The best case is when the value of RCA2 is 1. This means that the observed product is not imported.

RESULTS AND DISCUSSION

As far as the global production of soybean is concerned, in the observed time period there was an increase both in the area on which the legume was grown and in the production. The areas on which the soybean was produced in 2009 were 97 million hectares (ha). In 2018, the areas on which the soybean was produced were 124 million hectares (ha). This shows an increase of 27.8%. There was even bigger increase in production of 59%. In 2009 the production was 212 million of tones and in 2018 it was 337 million tons (source: FAO database). The data of the area on which soybean was grown, production and yield for the Republic of Serbia in the observed time period are given in Table 1.

Table 1. Area under soybean, production and yield from 2009 to 2018 in Serbia

Year	Area (ha)	Production (t)	Yield (t/ha)
2009	144386	349193	2.4
2010	170255	540859	3.2
2011	165253	440847	2.7
2012	162714	280638	1.7
2013	159724	385214	2.4
2014	154249	545898	3.5
2015	184841	454431	2.5
2016	182362	576446	3.2
2017	201712	461272	2.3
2018	196472	645607	3.3

Source: <http://www.stat.gov.rs>

In Table 1 it can be seen that the area on which soybean was grown increased by 3.48% annually. The production increased as well, and the annual growth rate was 7.07%. The average yield also increased, but as it was also the case with other crops grown in the Republic of Serbia due to lack of irrigation, it varied in accordance with climates. The input data for calculating the three previously mentioned indexes are from FAO database. The data were chosen in this way in order to allow the analysis of the competitiveness of export of soybean on both national and global market in accordance with SITC Revision 4. The competitiveness is the ability of domestic companies to sell their product to clients in the country and in foreign countries (Porter, 1990). The evidence of the good competitive position of Serbia is based on the fact that out of 10 observed years, this country had a positive foreign trade balance in 7 years as far as this area of economy is concerned. This means that the export of soybean was higher than the import in the same time period, as it is shown in Table 2.

Table 2. Foreign trade characteristics of soybean in Serbia

Year	Import (thousands USD)	Export (thousands USD)	Foreign trade exchange (thousands USD)	Foreign trade balance (thousands USD)
2009	20517	1843	22360	-18674
2010	18426	22692	41118	4266
2011	2458	20730	23188	18272
2012	29594	8059	37653	-21535
2013	10507	14100	24607	3593
2014	12517	9474	21991	-3043
2015	37874	55753	93627	17879
2016	4539	52237	56776	47698
2017	37717	49699	87416	11982
2018	7609	31030	38639	23421

Source: Compiled from FAOstat, World Bank, Statistical Office of the Republic of Serbia and UNCTADstat

Bearing in mind the aim of the research, a question must be raised whether soybean produced in Serbia is competitive on the international market. Further analysis demands the review of indexes of competitiveness. Table 3 presents the results of calculation for three above mentioned indexes: RCA, RCA1 and RCA2.

Table 3. Indexes of competitiveness of the Republic of Serbia from 2009 to 2018

Year	RCA	RCA1	RCA2
2009	-1.340	1.919	-0.835
2010	-0.254	2.943	0.104
2011	0.535	2.844	0.788
2012	-0.960	2.395	-0.572
2013	-0.354	2.501	0.146
2014	-0.570	2.310	-0.138
2015	-0.307	3.127	0.191
2016	0.572	3.027	0.840
2017	-0.344	2.947	0.137
2018	0.170	2.722	0.606
Period's average value	-0.285	2.673	0.127

Source: Compiled from FAOstat, World Bank, ITC, Statistical Office of the Republic of Serbia and UNCTADstat

In Table 3 it can be seen that the average value of RCA index is negative, and its value is -0.285. In the observed time period, except for years 2011, 2016 and 2018, the value of the index was negative. The negative value of the index shows that there were no competitive advantages on the national market. On the other hand, if the value of the index RCA1 is observed, it can be seen that its average value is 2.673 which shows high competitiveness of the soybean produced in Serbia on the international market. Moreover, the annual increase of value of this index shows that Serbia is competitive on the international market. The reason for this is the increasing numbers of countries, especially the ones from the west Europe, which are in demand of soybean of known origin which is not genetically modified. The positive average value of the Index of Net Business Performance (RCA2=0.127) shows that the soybean industry has a positive contribution rate on the foreign trade balance of the economy in the Republic of Serbia. The demand for soybean which is not genetically modified is increasing. This is the main reason for the good position of Serbia on the international market of soybean industry.

On the other hand, according to Živkov et al. (2016), the introduction and implementation of certain measures will improve and consolidate the position of the Republic of Serbia on the international market. The access to new technology, and investment in the basic means for production as well as further education of the producers would increase the yield. These measures would also allow for additional production for export. Additional production of organic soybean and its products with higher value per unit of product, which will be even more in demand in future, can have even more positive impact on the export of soybean.

CONCLUSION

In order to develop the long-term competitiveness, soybean industry cannot be based entirely on the maximal exploitation of the natural resources, but it also requires diversification that will allow further processing and production of high value-added products. The analysis of competitiveness of soybean industry shows that there is a competitive advantage on the national and international market. It is evident that there is potential for development of the soybean industry, which will help Serbia's economy in having sustainable growth in the future, as well as balanced foreign trade. On the basis of all calculated indexes, this research has shown the existence of the positive trend of increasing the export of soybean. However, the Republic of Serbia has to continue to indicate the origin of soybean, especially the one which is not genetically modified. Processing facilities must develop in order to produce high value-added products for the export. The soybean industry has modest impact on the foreign trade balance of the Republic of Serbia. However, if the previously mentioned measures are applied, the impact can be significantly higher in the future.

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DETECTION OF PHYTOPLASMAS IN FOREST SPECIES IN REPUBLIC OF SRPSKA

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Summary: *Phytoplasmas are associated with diseases spread throughout the world to hundreds of plant species. The main aim of the study was to check presence of phytoplasmas in forest trees, in urban and rural area in several location of Republic of Srpska. Plant samples from 16 plant genera were collected from five locations and subjected to nested PCR/RFLP analyses. Positive PCR products were sent to sequencing and sequence analyses were performed to double check identity of results. Six out of 42 plant samples were positive for presence of phytoplasma in nested PCR. The RFLP and sequence analyses reveled phytoplasmas from 16SrI group (aster yellows) and 16SrV group (elm yellows).*

Key words: *aster yellows, 16SrI, elm yellows, 16SrV, nested PCR, RFLP*

INTRODUCTION

Phytoplasmas are obligatory plant pathogens and, except in plants, can be maintained and propagated in insect hemolymph. With their destructive effect, many economically significant plants, both in our country and in the world, are endangered (Radulović *et al.*, 2018). Except in cultivated plants, host of phytoplasmas are species from the spontaneous flora found in and around the plantations (Mehle *et al.*, 2019). Phytoplasmas are transmitted by insect vectors from the families *Cicadellidae*, *Fulgoridae* and *Psyllidae* which feed on the phloem of the infected plant. Insects transmit phytoplasmas on persistently-propagatively manner, after inquisition phytoplasmas multiply in the body of the insect and the vector is infected all its life (Weintraub and Beanland, 2006).

Because forest species can represent reservoirs of phytoplasmas their detection is very important. Therefore, the main objective of this study was to analyze some forest species that can be potential reservoirs for phytoplasmas in the area of Republic of Srpska, Bosnia and Herzegovina, BiH. This review would provide information on what forest species could be potential hosts of phytoplasmas, pathogens of cultivated species. This research was conducted through activities related to the Euphresco project "Study on the diversity of phytoplasmas detected in European forests (PhyFor)".

MATERIAL AND METHODS

Surveys and samples collection

Samples were taken from beginig of June till last week of July in 2019. Five locations were at our disposal, three of which were in the Banja Luka area, one in the Gradiška area and one in the Kostajnica area. The first samples were taken from the University Campus in Banja Luka, 23 samples. One sample was taken from the St. Sava Alley, three from the Gradiška area, six from Kostajnica and nine from the Trešnjik resort in Banjaluka, which is a total of 42 samples from sixteen plant genera, *Populus* sp., *Platanus* sp., *Paulownia* sp., *Ulmus* sp., *Morus* sp., *Betula* sp., *Fraxinus* sp., *Quercus* sp., *Acer* sp., *Salix* sp., *Robinia* sp., *Carpinus* sp., *Sambucus* sp., *Tilia* sp., *Alnus* sp. and *Ostrya* sp.. In addition to trees that had symptoms, samples were also taken from asymptomatic trees. Samples were taken from leaves that were within easy reach of the accessible sides of the tree. The leaves were stored in the refrigerator, in bags with sample codes, for up to two days before removing the leaf midribs.

DNA extraction

A central leaf midrib with as little leaf tissue as possible was taken during sample preparation. The extracted tissue then was grinded with liquid-nitrogen in mortar with pestle until a powder-like plant sample was obtained. The plant samples prepared in this way were stored in tubes with sample codes in the freezer at -80°C. DNA isolation was performed according to the protocol described by Green *et al.* (1999).

PCR and RFLP analysis

A nested polymerase chain reaction (nested-PCR) was used to detect phytoplasmas. For this purpose, primer pair P1/P7 (Deng and Hiruki 1991; Schneider *et al.*, 1995) was used in direct PCR while in nested the obtained PCR product was used as a matrix for amplification with the primers R16F2n/R2 (Gundersen and Lee, 1996). The reaction was carried out in a 20 µl PCR mixture in a PCR thermocycler Applied Biosystems 2720. Fragment visualization was performed in a 1% agarose gel immersed in 1xTAE buffer at 100V. Positive PCR products obtained in nested PCR with R16F2n/R2 primers were subjected to Restriction Fragment Lenght Polymorphism (RFLP) analysis. The total reaction volume was 20 µl of RFLP mixture where PCR products were subjected to restriction with the *RsaI* endonuclease enzyme at 37 ° C for according to the manufacturer recommendation. After incubation, visualization was performed in 2.5% MetaPhor agarose (Lonza, Italy) gel immersed in 1xTBE buffer at 75V for 2 hours. All positive PCR products were sent for custom sequencing in both direction to Macrogen Europe. Sequences of PCR products were assembled using BioEdit software (Hall, 2011). A database search of homologous sequences was performed by BLAST analysis at NCBI (<http://ncbi.nlm.nih.gov/BLAST>). Gene sequences, 16S rRNA, were aligned with phytoplasma group/subgroup representatives available in GenBank using

Clustal W (Thompson *et al.*, 1994). All obtained sequences were stored in GenBank and accession numbers were obtained.

RESULTS AND DISCUSSION

Surveys and symptoms observed

Majority of observed plants did not show typical phytoplasma symptoms. On *Populus nigra* cv. *italica* - Lombardy poplar pronounced drying of the branches, especially in the lower part of the tree was recorded. No new shoots grew from the lower part of the tree, as is generally the case with this plant species. The foliage was spared, and the plant had a reduced growth (Figure 1).

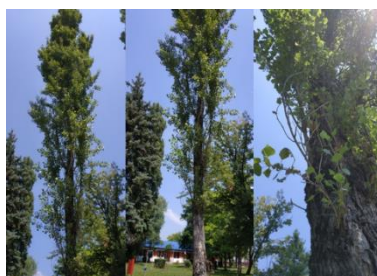


Figure 1. *Populus nigra* cv. *italica* – Lombardy poplar (defoliation on the lower branches, dieback, decrease in growth)

Ulmus sp. – elm from Gradiška locality that showed symptoms of drying. In the case of less infected leaves, the rim turned brown, while the older leaves were completely necrotic. The infection spread from the edge to the center of the leaf. V-shaped drying was present on certain leaves. The leaves curled upwards. A decay was observed (Figure 2). The campus elm sample did not show specific EY symptoms.



Figure 2. *Ulmus* sp. - elm (symptoms of necrosis on the leaves, spreading from the edges to the centre and twisting upwards)

On location Trešnjik (Banja Luka) phytoplasma - like symptoms were observed only on *Acer negundo* – boxelder, where the leaves dried, twisted down and deformed (Figure 3).



Figure 3. *Acer negundo* (drying leaves, twisting down)

Molecular analyses

Nested PCR resulted the presence of PCR products of about 1250bp in length in 6 out of 42 tested samples. RFLP with the *RsaI* enzyme revealed the presence of phytoplasmas of the 16SrI and 16SrV ribosomal groups (Table 1, Figure 4). The obtained results of the RFPL profiles were compared with the reference profiles for the *RsaI* restriction enzyme that was used reference (Lee *et al.*, 1998).

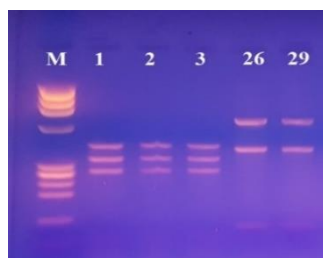


Figure 4. Electrophoresis on 2.5% MetaPhor agarose gel RFLP products with *RsaI* restriction enzyme. M- marker (φx174 DNA / BsuRI (HaeIII) (Fermentas, Lithuania)). 1, 2, 3, 26, 29 are positive samples.

Comparison of 1.25kb of R16F2n/R2 primed sequence of 16S rDNA of three phytoplasma isolates from black poplar (sample 1 and 2) and plane tree. (sample 3) (Figure 4) to other phytoplasma sequences reported in GenBank using the BLAST tool, revealed that samples had maximum identity of 99,6 to 99,8% to the phytoplasma associated with *Populus nigra* - black poplar in Croatia (JN005706). Comparison of nucleotide sequences from elm (sample 26) and black alder (sample 29) (Figure 4) to other phytoplasma sequences indicates that the phytoplasma associated with elm had maximum identity, 100% with sequences of 'Candidatus Phytoplasma ulmi' elm isolates from Belgium (MH583020), United Kingdom (KT381877) and Serbia (HM038460), while phytoplasma sequence from alder showed identity of 100% to 'flavescence dorée' isolate from grapevine from Serbia (EF581165).

Table 1. Phytoplasma groups detected in positive samples

Plant species	Location	Phytoplasma	Accession numbers
<i>Populus nigra</i> cv. <i>Italica</i>	Campus	16SrI	MN662619
<i>Populus nigra</i> cv. <i>Italica</i>	Campus	16SrI	MN662620
<i>Platanus acerifolia</i>	Campus	16SrI	MN662621
<i>Ulmus</i> sp.	Gradiška	16SrV	MN662622
<i>Alnus glutinosa</i>	Kostajnica	16SrV	MN662623
<i>Acer pseudoplatanus</i>	Trešnjik	-	-

On species *Populus nigra* cv. *italica* – Lombardy poplar and *Platanus acerifolia* – plane tree the presence of phytoplasmas from the 16SrI (Aster yellows) group was detected. 'Ca. P. asteris' is associated with symptoms such as yellowing of the plants, sterility of the flowers, malformations of the organs and lower plant quality. It attacks over 300 plant species, the most important being lettuce, carrots, celery and more (Mahr, 2015). Phytoplasmas from 16SrI (Aster yellows) group are in association with Tomato big bud, various aster yellows diseases, Mitsuba witches' broom, Eggplant dwarf, Maize bushy stunt, Mulberry dwarf, Strawberry green petal, Paulownia witches' broom, Blueberry stunt and Apricot chlorotic leaf roll. The host plants of the phytoplasmas from this group are plants of the *Asteraceae* family: tomatoes, zucchini, spinach, onions, hemp, eggplant, corn, mulberry, strawberry, peacock, blueberry and peach (Lee *et al.*, 1998; Mahr, 2015).

In the detection of phytoplasmas in Germany, *Ulmus* spp. rarely showed symptoms. Chlorosis and lagging growth have been observed (Schneider *et al.*, 2019). The observed symptoms, leaf yellowing, drying and premature decline were observed in infected plants with 'Ca. P. ulmi'. In Croatia on *Ulmus* spp. leaf yellowing, drying, early leaf fall and branch drying were noted (Katanić *et al.*, 2016). 'Ca. P. ulmi' is a quarantine regulated pest and is on the EPPO A1 list and EU Annex I / A2 list (EPPO, 2019). The 16SrV (Elm yellows) group contains phytoplasmas in association with Elm yellows, Cherry lethal yellows, Alder yellows, Sweet potato little leaf, Peach yellow leaf roll, Rubus stunt, Eucalyptus little leaf and 'flavescence doree' (Lee *et al.*, 1998). The grapevine is the most abundant, while the other delicate plants are: cherry, peach, batat, eucalyptus and species of the genus *Rubus*. Insects associated with the possible transfer of phytoplasmas from this group are *Scaphoideus luteolus*, *Philaenus spumarius*, *Allygus atomarius*, *Oncopsis alni* (Carraro *et al.* 2004; Marjanovic *et al.*, 2019).

FD-related strains were found in *Alnus glutinosa* in Slovenia, in hazelnut molecular characterization confirmed that the strains found were identical to those associated with the 'flavescence doree' found in grapevine in 2005 (Mehle *et al.*, 2019). *Alnus* sp. infected with phytoplasma from the 16SrV group was also found in Gradiška area (Radulović *et al.*, 2018). In a study in Serbia, 100% of *Alnus glutinosa* plant specimens were positive for the presence of phytoplasmas from the 16SrV group. Also, 100% infection was reported in Germany (Holz *et al.*, 2016; Marjanovic *et al.*, 2019). AldY is in the same group as 'flavescence doree' (FD)

and Palatinate grapevine yellows (PGY). It is thought that FD and PGY could originate from ALdY as a consequence of improper transfer from *Alder* sp. on the vine. The same can be considered for elm yellows (EY) phytoplasma. Even if EY phytoplasma does not pose a threat to elms, it can be a starting point for the development of phytoplasmas that would transfer to more sensitive species (Arnaud *et al.* 2007; Katanić *et al.* 2016). '*Flavescence dorée*' is a quarantine regulated pest and is on the EPPO A2 list and EU Annex II/A2 and Annex II/B. There is evidence that ALdY strains that infect alder trees in the Palatinate region of Germany can also infect grapevines in nearby vineyards and cause a disease called Palatinate Grapevine (PGY). Because of this, infected alders can serve as inoculum sources. Also, due to the close phylogenetic relationship of this phytoplasma and '*flavescence doree*', these trees can serve as inoculum for FD where present (Marccone, 2015; Marjanovic *et al.*, 2019).

In the study phytoplasma was detected *A. pseudoplatanus* but the characterisation hasn't done yet. Generally *A. negundo* was positive the presence of phytoplasmas from the 16SrI group in the urban area of Warsaw in Poland. Symptoms such as decreased apical growth and changes in leaf size and color on older trees were observed. In the later stages, leaf defoliation and necrosis occurred (Kamińska and Śliwa, 2005). Similar symptoms were also observed on the *Acer negundo* from resort Tresnjik but no phytoplasma was detected. Also, research in Poland indicates that *A. pseudoplatanus*, which, due to its low concentration, did not succumb to RFLP, could be in the 16SrI - aster yellows group. Phytoplasmas are quite unevenly distributed in the plants and their concentration is very low so to improve detection potential root samples will be included in the future research.

CONCLUSION

- Surveys for phytoplasmas in the forest species were conducted for the first time in Republic of Srpska, BiH;
- On plant species *Populus nigra* cv. *italica*, *Platanus acerifolia*, *Ulmus* sp., *Alnus glutinosa* and *Acer pseudoplatanus* presence of phytoplasma was detected;
- *Populus nigra* cv. *italica* and *Platanus acerifolia* had maximum identity of 99,6 to 99,8% to the phytoplasma associated with '*Populus nigra*' phytoplasma from black poplar in Croatia;
- *Ulmus* sp. had maximum identity of 100% with sequence of '*Candidatus Phytoplasma ulmi*' from Belgium, United Kingdom and Serbia, while phytoplasma sequence from *Alnus glutinosa* showed identity of 100% to FD isolate of grapevine from Serbia;
- In the species *Acer pseudoplatanus* phytoplasma is detected, but due to low concentration characterization has not been completed;
- It is necessary to study the surrounding flora and fauna (insects) to create a clearer picture of the distribution and importance of phytoplasmas in Bosnia and Herzegovina.

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ENDOCRINE DISRUPTORS IN PLASTICS: MARIBOR RESIDENTS' AWARENESS OF WATER QUALITY FROM TAPS AND PLASTIC BOTTLES

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Summary: *Research on endocrine disruptors and their effects on health has been increasing. Recently attention has been given to the migration of endocrine disruptors from packaging into food. This packaging includes plastic bottles for water storage. This study mainly investigated Maribor residents' level of awareness of endocrine disruptors. A questionnaire was completed by 93 Maribor residents. The research focused on their past experience with hormonal disruptors and collected some important information and opinions about their knowledge of symptoms caused by hormonal disruptors. The study determined what Maribor citizens think about drinking water quality in Maribor and why some people prefer bottled water over drinking fountains.*

Keywords: *hormonal disruptors, plastic bottles, water fountains, public awareness*

INTRODUCTION

The European Food Safety Authority defines hormone or endocrine disruptors as substances that can interact with normal hormonal action (EFSA 2016). Potential endocrine disruptors are mostly manmade and are found in various materials, such as pesticides, metals, or additives in food and products for personal care (WHO 2016).

The most popular material for food and beverage packaging is polyethylene terephthalate (PET). PET bottles are disposable single-use bottles (Polajžer 2012). They contain acetaldehyde and bisphenol A (BPA), which can be released into water by high temperatures (NIEHS 2008; Fan et al. 2014).

BPA can mimic the mechanism of the hormone estrogen and has an impact on reproduction (Matuszczak et al. 2019). Infants and children exposed to BPA may express behavioral and developmental disorders (Johnson et al. 2015).

The main goal of this study was to investigate Maribor residents' level of awareness of hormone disruptors and their presence in everyday life. The research focused on their experience with endocrine disruptors, the main groups of these chemicals, their health impact, and their presence in plastic (PET) bottles. The study investigated what Maribor residents think about the quality of drinking water from taps, public water fountains, and plastic bottles.

MATERIAL AND METHODS

A survey was conducted from the beginning of June to the end of September 2015. The questionnaire was published online at 1KA. Questionnaires were also given to individuals encountered near public water fountains at the University of Maribor's Faculty of Agriculture and Life Sciences and Faculty of Medicine, and at Maribor University Medical Center. The questionnaire consisted of 31 questions. The first part of the questionnaire contained sociodemographic questions about respondents' sex, age, level of education, and profession. The second part of the questionnaire included questions about knowledge of the endocrine system, hormonal disruptors, and their impact on health, and the third part included questions about the quality of tap water, water from public fountains, and water from plastic bottles.

The survey was completed by 93 respondents (N=93). The sample included males and females living in Maribor. All age levels (up to 20 years old, 21–40 years old, 41–80 years old, and 61 years old or older) and all levels of education (primary school, secondary school, bachelor's degree, master's degree, and doctorate) were represented. The study also took into account respondents' professions, which were divided into groups (non-healthcare professions, healthcare professions, the unemployed, and none indicated). The research sample was not obtained randomly, and so it is rather small. Consequently, the representativeness of the results is poor regarding certain responses. It is important to recognize that the actual situation may deviate from the analysis of some of the answers presented. It is anticipated that a larger research sample would yield more reliable results, contributing to knowledge of public awareness.

The results of the survey were first presented with contingency tables showing response frequencies and associated percentages. Microsoft Office Excel 2013 was used to draw pie charts and bar charts from the tables provided. In statistics, the chi-square test is often used to correlate two descriptive variables, but it assumes that all expected frequencies are at least five (Košmelj 2007; Cochran 1954). SPSS, version 20.0, was used to analyze correlations between descriptive variables. In most cases, it turned out that the assumption above was not fulfilled. In the literature, the authors propose combining categories if meaningful (Košmelj 2007). For small samples, Fisher's exact test may be used, but it does not require the above assumptions. The test was performed using an online calculator (Exact $r^x c$ Contingency Table 2016).

RESULTS AND DISCUSSION

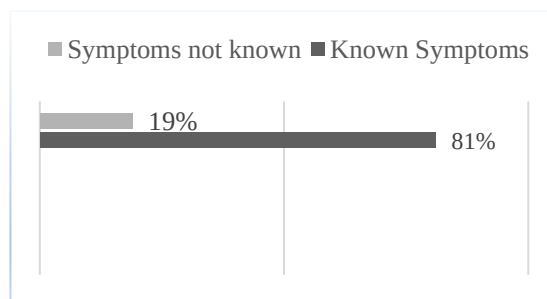


Fig. 1 Are you familiar with symptoms associated with endocrine disorders ?

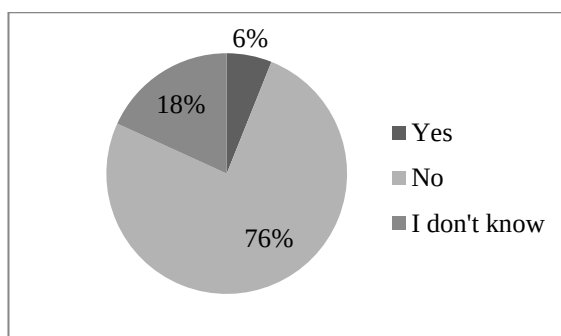


Fig. 2 The PET bottles purchased in stores are safe and contain no substances harmful to health

CONCLUSION

In the last few years, much research has been carried out on hormone disruptors, confirming that hormone disruptors are a very important environmental and health problem (Skalicky 2012). This study has shown that Maribor residents have poor knowledge of hormone disruptors. One of the main reasons is the media, which provide insufficient information about hormone disruptors. Educational institutions also do not teach enough about such chemicals. A study by Medved (2015) showed that students from the University of Maribor's Faculty of Medicine are most aware of hormonal disruptors. Forty percent of students from the Faculty of Agriculture had never heard of endocrine disruptors. Medical students have most often heard about endocrine disruptors in class (48%). Agricultural students have heard about endocrine disruptors online (36%).

In most cases, education and age do not appear to correlate with knowledge about hormone disruptors and water quality. It is difficult for people to avoid water stored in PET bottles. Because such bottles are very convenient, people use them daily. Although they are practical, people are unaware of the harmful chemicals they release into the water. When there is no tap water or public drinking fountain available, it is advisable to use packaging made from safe materials (steel or glass).

The quality of bottled water can be compared to the quality of tap water. Unlike tap water, bottled water is in contact with plastic for a long time. Plastics can release various substances into the water. These may be residues of materials and additives, as well as reaction and decomposition products (NIPH 2015).

Eighty-one percent of respondents are aware of the symptoms resulting from endocrine disruptors. Nineteen percent of respondents do not know the symptoms. Symptoms associated with endocrine disorders are feminization of males and masculinization of females, behavioral disorders, hypospadias, and thyroid disorders.

Seventy-six percent of respondents were of the opinion that the plastic bottles purchased in stores are not safe to use and that they contain substances harmful to health; only 6% disagreed, and the remaining 18% did not respond to the question. Statistical analysis shows that both respondents' age ($p = 0.599$) and education ($p = 0.263$) do not affect the opinion that plastic bottles purchased in stores are unsafe to use and that they contain substances that can harm health.

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BLOOD HAEMATOLOGICAL AND BIOCHEMICAL PARAMETERS AND LEAD CONCENTRATION IN GRIFFON VULTURES (*GYPS FULVUS*) IN CROATIA

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Summary: *Haematological and biochemical analysis and blood lead concentration is very essential field in veterinary medicine that provides inexpensive and reliable results in determinating animal health. Lead is considered a highly toxic contaminant with important impact to birds of prey. The study was realized in all admitted Griffon vultures at the Beli Visitor and Rescue Centre for Griffon vultures in Croatia. The goal of this study was to investigate haematological and biochemical parameters and lead concentration in Griffon vultures in Croatia. In order to investigate haematological and biochemical profiles and lead concentration of birds of prey, the CBC, haemoglobin concentration, haematocrit, blood cell indices, total WBC and lead concentration of 12 Griffon vultures (*Gyps fulvus*) were analysed. Obtained results provide essential and play a very important role for evaluating the health status and pathologic conditions of these birds of prey. Since data of haematological and biochemical values and lead concentration of these birds of prey are still scarce, further studies are still needed to assess the physiological range of haematological, biochemical and lead values in Griffon Vultures in Croatia.*

Key words: *haematology, biochemistry, lead, vulture, blood*

INTRODUCTION

The Griffon vulture (*Gyps fulvus*) is one of the largest bird species in the world and endangered bird of prey in Croatia and is at very high risk of extinction. Blood haematological and biochemical parameters and toxicological (especially lead) concentration are very important diagnostic tools in determining the general health condition in these animals. To assess health status and pathological conditions of vultures in Croatia, knowledge of their normal values of blood parameters and lead concentration is of great importance. The goal of this study was to investigate haematological and biochemical parameters and lead concentration in Griffon vultures in Croatia.

MATERIAL AND METHODS

In this research blood of twelve Griffon Vultures were included. Blood samples were collected by veterinarian by puncturing the radial vein, immediately upon arrival of each individual. All blood samples were stored in tubes with anticoagulant for hematology and

toxicology and without anticoagulant for biochemical analyses. In collected samples, complete blood count (CBC), serum biochemical panel and lead concentration were analysed. Blood chemistry profiles and lead concentration were determined within 2 days by automatic analyser. Complete blood count was determined within 5 hours after collecting blood samples.

RESULTS AND DISCUSSION

In our study, we found a considerable range in the values of some parameters, e.g. PCV ranged from 32 – 52%, haemoglobin (4,2 – 14,97 g/dl), MCV (155,34 – 221,05 fL), MCH (20,3 – 64,79 pg), MCHC (12,73 – 29,35 g/dl), total WBC ($6,8 - 24,1 \times 10^3 /\mu\text{L}$), heterophils ($2,72 - 19,76 \times 10^3 /\mu\text{L}$), lymphocytes ($0,33 - 3,35 \times 10^3 /\mu\text{L}$), glucose (13,8 – 19,8 mmol/L), total proteins (25 – 50 g/L), GLOB (17 – 37 g/L), CHOL (3,4 – 10,1 mmol/L), UREA (0,5 – 3,7 mmol/L). Liver enzymes – AST ranged from 184 – 355 U/L, ALT 21 – 55 U/L and ALKP 25 – 317 U/L. CK ranged from 144 – 491 U/L, LDH 122 – 453 U/L. Lead concentration ranged from $<0,01 - 0,546$ ppm.

After admitting blood samples we found a considerable variation in some parameters which can suggest pathological conditions in some individuals. Further investigation of blood samples after rehabilitation and treatment of individuals should be obtained and compare with results before rehabilitation.

CONCLUSION

Obtained results provide essential and play a very important role for evaluating the health status and pathologic conditions of these birds of pray. Since data of haematological and biochemical values and lead concentration of these birds of pray are still scarce, further studies are still needed to assess the physiological range of haematological, biochemical and lead values in Griffon Vultures in Croatia.

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HISTOLOGICAL EVALUATION OF ACHILLES TENDON HEALING IN RABBITS

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Summary: *The Achilles tendon in rabbits is found, as in other species, on the back of the lower leg. It's made of tendons of three muscles: musculus gastrocnemius medialis, musculus gastrocnemius lateralis and musculus soleus. Not all parts of tendon have the same vascularisation properties. Therefore different segments of tendon should not heal equally. The goal of this study is to evaluate healing properties of different segments of Achilles tendon in rabbit model. 12 rabbits were used as a model because of their anatomic similarity with the tendon in humans and because of their calm nature and accessibility. The research was carried on rabbits of both sex, 6 males and 6 females, around three kilograms of bodyweight and one year old. In this study a Bonar scale was used to compare histological structures of scarificated tendons and intact ones. The result of histological examination confirms that the fastest and the best healing is in the part of the tendon attached to the muscle, then the part attached to the heel bone, and the slowest healing is in the middle of the tendon.*

Key words: *Achilles tendon, histology, healing, rabbits*

INTRODUCTION

Tendons in general are made out of connective tissue, mostly collagen fibers, tenocytes that are cellular elements and intercellular substances that consists of glycosaminoglycan which has a high affinity for water binding. This characteristic of this protein is highly important in tendons because of the high percentage of water that is found in them (55-70%). All tendons are covered with sheath under which is a joint-like cavity filled with fluid. The Achilles tendon in rabbits is found, as in other species, on the back of the lower leg. It's made of tendons of three muscles: *musculus gastrocnemius medialis*, *musculus gastrocnemius lateralis* and *musculus soleus*. The tendon does not

have a real sheath, but it's lined with a thin sheath that we call paratenonium. Paratenonium is described as a fat connective tissue or as a vascular connective tissue rich in elastic fibers. Blood in tendons is supplied through three main compounds: the internal system at the junction of the muscle and tendon, at the junction of the tendon and bone, and by the external system via the paratenonium or synovial sheath. Not all parts of tendon have the same vascularisation properties. Therefore different segments of tendon should not heal equally. The goal of this study is to evaluate healing properties of different segments of Achilles tendon in rabbit model.

Tendon injuries represent a major problem during the treatment. Today we know that there are three stages of healing if we look in the time aspect. The first period is the period of inflammatory response and this one is the most important in the first days, then we have period characterized by fibrillogenesis and fibriloplasia, and the last one is marked with progressive alignment and organization of collagen fibers. The ability of the tendon to heal is intrinsic and extrinsic. In healing of tendons, important role is creating of attachments. They are used to increase blood flow from the surrounding tissues. It takes around three weeks for them to create. Also, mechanical loading after tendon injury accelerates tenocytes metabolism and increase their production of collagen and the extracellular matrix in animal models.

This research was approved by Ministry of Agriculture and Ethic Committee of Faculty of Veterinary Medicine University of Zagreb.

MATERIALS AND METHODS

In this research, 12 rabbits were used as a model because of their anatomic similarity with the tendon in humans and because of their calm nature and accessibility. The research was carried on rabbits of both sex, 6 males and 6 females, around three kilograms of bodyweight and one year old. In 8 rabbits all Achilles tendons were used as a inflammation model. Intact tendons of 4 rabbits were used as a model of normal histological structure of the tendon. During the research rabbits were located at the Clinic for Surgery, Orthopaedic and Ophthalmology. The rabbits were sedated with the intramuscular injection with the combination of ketamine and xylazine and as the induction in the anesthesia, propofol was administrated iv. For additional analgesia 2% Lidocaine solution was used epidurally. The approach on the tendon was made on the posteromedial side of the leg (Fig.1.).



Fig.1. Posteromedial approach and longitudinal tenotomy of the Achilles tendon using blade No 15. (Pećin, 2014.)

The procedure began with resection of the tendon sheath (paratenonium). Scarification and tendotomy of 16 tendons were then made with three longitudinal cuts from the attachment of the muscle all the way to the calcaneus using blade No 15. (BBraun, Tuttlingen, Germany). This incision was made to cause inflammation in the tendon (Friedrich and al. 2011). During the whole surgical procedure, the tendons were kept wet with cristaloid solution, because of their tendency to dehidrate. The paratendon was reconstructed using running suture with pds 4-0, and skin was then closed in running fashion using dermalon 4-0. Eight intact tendons (in 4 rabbits total) were used to determine normal histological structure of Achilles tendon in rabbits. Analgesia before and after the surgery was achieved with butorfanol (Torbugesic, Fort Dodge, Iowa, Usa) as it does not affect tendon healing like NSAID's do (SIGBJORN, 2011). After 4 weeks rabbits were euthanized using T-61 administered iv after im sedation with ketamine and xylazine. Achilles tendons were harvested in full lenght from insertion with muscles to attachement on calcaneus for histological examination. The goal of the histological examination was to define the number and the layout of elastic fibers, number of tenocytes and fibroblast, the amount of glucosaminoglycan of extracellular matrix, position and direction of collagen fibers in tendons. Each tendon was cut lengthwise in two parts and then each in three equal parts, 1 centimeter long (Fig.2.).



Figure 2. Achilles tendon cut lengthwise, each segment one centimeter long, (Pećin, 2014.)

In this study a Bonar scale was used to compare histological structures of scarificated tendons and intact ones. Same scale was used in previous published studies for histological evaluation of tendon healing (MAFFULLI et al. 2008). Bonar score scale uses grade system from 0-3 for evaluation of tendon healing (Fig.3.).

Variables	Grade 0	Grade 1	Grade 2	Grade 3
Tenocytes	Inconspicuous elongated spindle shaped nuclei with no obvious cytoplasm at light microscopy	Increased roundness: nucleus becomes more ovoid to round in shape without conspicuous cytoplasm	Increased roundness and size: the nucleus is round, slightly enlarged and a small amount of cytoplasm is visible	Nucleus is round, large with abundant cytoplasm and lacuna formation (chondroid change)
Ground substance	No stainable ground substance	Stainable mucin between fibers but bundles still discrete	Stainable mucin between fibers with loss of clear demarcation of bundles	Abundant mucin throughout with inconspicuous collagen staining
Collagen	Collagen arranged in tightly cohesive well-demarcated bundles with a smooth dense bright homogeneous polarization pattern with normal crimping	Diminished fiber polarization: separation of individual fibers with maintenance of demarcated bundles	Bundle changes: separation of fibers with loss of demarcation of bundles giving rise to expansion of the tissue overall and clear loss of normal polarization pattern	Marked separation of fibers with complete loss of architecture
Vascularity	Inconspicuous blood vessels coursing between bundles	Occasional cluster of capillaries, less than one per 10 high-power fields	1-2 clusters of capillaries per 10 high power fields	Greater than two clusters per 10 high-power fields

Figure 3. Evaluation of tendon healing process and parameters observed with Bonar score scale, (Pećin, 2014.)

RESULTS AND DISCUSSION

The following histological changes were observed: cellularity, shape and size of the tenocyte nucleus, changes in the cytoplasm of cells, position, connectivity and orientation of collagen fibers, organization of bundles at longitudinal and cross section, amount of colored glycosaminoglycans of the basic substance, presence of elastic and reticulin fibers and the presence of blood vessels. Four weeks after the tenotomy, a significant difference was noticed in all three parts of the tendon between tendons undergoing tenotomy and intact tendons. In order to better assess the histological healing of tendons undergoing tenotomy, it is important to be familiar with the physiological histological layout of the tendon. The number of tenocytes in intact tendons is lower than in tenotomy. Collagen fibers are properly positioned and longitudinal, vacularization is present in all parts of the tendon, mostly in the part that is attached to the muscle, and decreases toward the heel bone. Elastic fibers are rarely present.

In order to understand healing, it is important to know how certain elements affect it. The healing process is greatly influenced by the number of tenocytes, so they are the parameter that was given the most notice. It has been observed that a greater number of tenocytes with less ground substance are present in the part of the tendon which is characterized by faster healing. Increased vascularization and newly formed blood vessels do not necessarily mean faster healing, since they do not function as true blood vessels, but greater presence in some parts of the tendon that has the best result in healing, has been noticed. Knowing the structure of the tendon, we know how tenocytes synthesize collagen. Considering that and the fact that we know the function of tenocytes in the healing process, we can say that the increased amount of collagen fibers affects healing speed.

Given that the Bonar scale ranges from 0-3, even small average differences in results represent significant differences in healing. As is shown in the table, number of tenocytes in the tendons after tenotomy, was the highest in the part attached to the muscle, and the lowest in the middle part. They are irregular in shape and scattered around. Ground substance was lowest in the part of the tendon that had the highest number of tenocytes. Collagen fibers are longitudinally laid, irregular and in various directions.(Fig.4) Vascularization is dense, especially in the part related to the muscle.(Fig. 5.)

Table 1. The sum of values of the Bonar scale to assess histological differences between groups. The 1,2,3 parts of the tendons are from proximal to the distal part of the sampling point(1-muscle part, 2-central part of the tendon, 3-part of the tendon attached to the muscle; SG-segment of the tendon, TN-tenocytes, GS-ground substance, CF-collagen fibers, VS-vascularization, IT-intact tendons, TT-tendons after tenotomy)

SG	TN	GS	CF	VS	
1	1,12	1,44	1,44	1,92	IT
2	1,00	1,49	1,55	1,58	
3	1,10	1,50	1,78	1,72	
1	1,72	1,93	1,90	2,00	T T
2	1,40	2,05	1,55	1,83	
3	1,68	2,00	1,78	1,89	

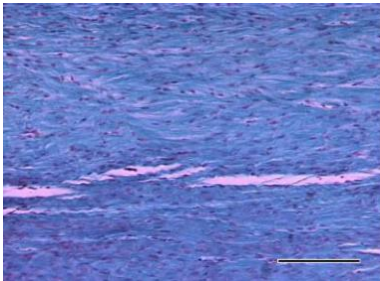


Figure 4. Irregular orientation of collagen fibers in the tendon after the tenotomy, Massone trithrome stain, x10, scale , 200 um (Pećin, 2014).

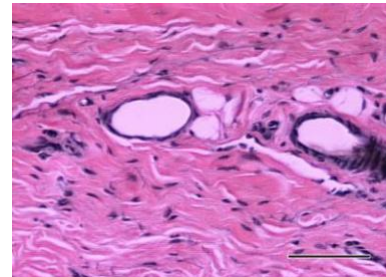


Figure 5. Vascularization in the tendon after tenotomy, Verhoeff-Van Gieson stain, x20, scale 100 um (Pećin, 2014)

CONCLUSION

The increase in tenocyte count is proportional to the less time it takes the tendon to heal. Also, if the part of the tendon has better vascularization, it has greater supply of nutrients, therefore it heals faster. All that parameters important for healing, were best expressed in the part of the tendon attached to the muscle. The result of histological examination confirms that the fastest and the best healing is in the part of the tendon attached to the muscle, then the part attached to the heel bone, and the slowest healing is in the middle of the tendon.

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VETERINARY STUDENTS' KNOWLEDGE AND PERCEPTIONS ABOUT ANTIMICROBIAL RESISTANCE AND ANTIMICROBIAL USE

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Summary: *The objective of this study was to investigate the the knowledge and perceptions about antimicrobial resistance (AMR) and antimicrobial use among veterinary students. In the study conducted in 2019, 46 forth year from agreed to finished anonymous questionnaire. Regarding the opinion of respondents whether antimicrobial use by veterinarians contributes to the overall burden of AMR, a slightly higher percentage of students chose moderate (43.48%) than strong contribution (39.13%). This study reveals that veterinary students from Novi Sad have a relatively good understanding of AMR.*

Key words: *student, knowledge, antimicrobial resistance, veterinary*

INTRODUCTION

Antimicrobial resistance (AMR) is one of the greatest challenges and the biggest threats currently facing human health and animal health and wellbeing, contributing to up to 25,000 deaths of humans in Europe every year [Marston et al., 2016]. Some newly emerging resistant bacteria in animals are transmitted to human, mainly via meat and other foods of animal origin or through direct contact with farm animals [WHO, 2002].

Irrational use of antibiotics is directly associated with AMR. Numerous interventions have been proposed to restrain AMR. Training of all healthcare professionals is highlighted in the WHO's Global Action Plan on AMR to combat this emerging public health problem [WHO, 2015]. Furthermore, Alothman et al. [2016] showed that physicians consider education to be among the most effective tools for reducing the problem of AMR and improving prescribing practices. Future veterinarians, as well as all healthcare professionals as prospective prescribers have to be educated and prepared to rationally manage the antimicrobial use.

Therefore, the present study is undertaken to investigated the knowledge and perceptions about AMR and antimicrobial use among veterinary students at University of Novi Sad in Serbia after they finished Pharmacology course.

MATERIAL AND METHODS

The study was conducted at University of Novi Sad between April and May of 2019 among 46 veterinary students from Faculty of Agriculture, Department of veterinary medicine. All

approached students were fourth year of the study and received teaching on antibiotics in their pharmacology course. A total of 46 anonymous questionnaires were distributed and all students agreed to complete it. The response rate was 100 %. The questionnaire used in this study is based on the questionnaire of Hardefeldt et al. [2018a]. Statistical analysis was performed using SPSS software version 23. Descriptive statistics were used to analyze the data. Chi-square test was used to compare the answers given by the two groups of students (female and male) and p values less than 0.05 were considered significant. The study received approval of Ethics Committee at Faculty of Agriculture, University of Novi Sad (approval number 9000/1160/2).

The questionnaire used in the study consisted of 4 sections with various questions in each section. In this study first and fourth section are used. The first section contained demographic questions such as age and gender structure of respondents, their graduating year, main area of interest and common opinion regarding veterinary antimicrobial use contributing to common antimicrobial resistance. The final section required the students to indicate their opinion on antimicrobial stewardship, lectures about the use of antimicrobials and also their knowledge of mechanisms of antimicrobial resistance.

RESULTS AND DISCUSSION

The results obtained from the study included sociodemographic characteristics of respondents. The average age of students was 23.5 years, while 54.35% of them were female and 45.65% male. More respondents (56.52%) expected to graduate in 2022. Almost 20% of students (19.57%) still haven't decided their main area of interest, whereas the highest number indicated companion animals which was almost 30% of all respondents (28.26%). All sociodemographic and academic characteristics of respondents are listed in Table 1.

Table 1. Sociodemographic and academic characteristics of veterinary students at University of Novi Sad

Characteristics	n	%
Age (years)		
22	3	6.52
23	36	78.26
24	6	13.04
25	0	0
26	1	2.17
Gender		
Male	21	45.65
Female	25	54.35
Graduating year		
2021	20	43.48
2022	26	56.52
Main area of interest		

Bovine	7	15.22
Public health, government, industry, research	5	10.87
Equine	4	8.70
Companion animals	13	28.26
Haven't decided yet	9	19.57
A mix of the above	8	17.39

Regarding the opinion of respondents whether antimicrobial use by veterinarians contributes to the overall burden of AMR, a slightly higher percentage of students chose moderate (43.48%) than strong contribution (39.13%). The rest of the respondents have shown poor knowledge about antimicrobial stewardship choosing not sure (13.04%) and no contribution (4.35%) (Table 2).

Table 2. Opinion of respondents indicating how much they think antimicrobial use by veterinarians contributes to the overall burden of antimicrobial resistance

Opinion of respondents	n	%
Not sure	6	13.04
No contribution	2	4.35
Moderate contribution	20	43.48
Strong contribution	18	39.13

Students reported having a good understanding of antimicrobial stewardship, but slightly more corrected answers (strongly agree and agree) were recorded among female (39.13%) than male students (15.22%). Almost the same number of both female and male students suggested that prudent antimicrobial use was taught less than was perceived necessary (41.3%), slightly more female students (21.74%). Most respondents responded having a good understanding of antimicrobial resistance mechanisms, but more female (47.83%) than male students were aware of this fact (36.96%). All results are presented in Table 3.

Table 3. Opinion of students regarding the teaching of antimicrobial resistance (AMR), antimicrobial use (AU) and antimicrobial stewardship (AMS).

	Strongly agree n f/m	Agree n f/m	Neither agree nor disagree n f/m	Disagree n f/m	Strongly disagree n f/m
I understand what AMS is	18/16	7/4	0	0/1	0
The amount of teaching time for prudent AU is about right	5/3	8/7	2/1	10/9	0/1
I understand AMR mechanisms	12/8	10/9	3/1	0/3	0

*f-female; m-male

DISCUSSION

Assessing knowledge and perceptions that guide antimicrobial stewardship among veterinary students represents an important part in counteracting the problem of antimicrobial resistance. There was only one survey conducted in Serbia among medical students regarding the teaching of prudent antibiotic prescribing [Pulcini et al., 2015], while one published study examined the general populations on this issue in our country [Horvat et al., 2017].

The term antimicrobial stewardship (AMS) is relatively new and represents an intergrated and multidisciplinary approach that optimize drug use, choice, dosing, duration, and rout of administration, while minimize the emergence of resistance and other adverse effects [Guardabassi and Prescott, 2015]. Actually, AMS does not mean stopping the use of antimicrobials, it involves the development of strategies to ensure that they are used effectively [Bowen, 2013]. Our findings suggest that students reported having a good understanding of antimicrobial stewardship, but slightly more corrected answers was given by female (54.35%) than male students (43.48%). Contrary to finding of our study, very low (17.1%) percentage of veterinarians in Nigeria have heard of AMS what is suggested that there is little or no AMS education awareness among them [Anyanwu et al, 2017].

Guidelines on prudent use of antimicrobial drugs have been developed as an integral part of veterinary practice and they are essential to preserve the effectiveness of these important drugs and consequentially to minimize the selection of resistant organism [WHO, 2000]. Present study revealed that almost the same number of both female and male students (43.48%) suggested that prudent antimicrobial use was taught less than was percieved necessary. Having appropriate knowledge on prudent antimicrobials use is important, since current and future veterinarians can play a crucial role in reducing the inappropriate use of antimicrobials.

In Serbia, veterinary medicine is a 6 years full-time University degree programme, courses on antibiotics and AMR are taught in the 3rd year of the study. Most respondents in our study responded having a good understanding of AMR, but more female (47.83%) than male (36.96%) students were aware of this fact. Similar to this study, veterinary students in Australia reported having a good understanding of the mechanisms of AMR, but their study did not assess the understanding of AMR mechanisms [Hardefeldt et al., 2018a].

In our study, almost half of respondents chose moderate (43.48%), while 39.13% of them chose strong contribution of antimicrobial use by veterinarians to the overall burden of AMR. In Australia, veterinarians frequently reported feeling partly responsible for AMR in human medicine and often were concerned about that contribution. Interestingly, although most companion animal veterinarians admitted to the overuse of antimicrobials, they thought that antimicrobial use in the dairy and intensive animal industries was most to blame, whereas veterinarians treating dairy cattle attributed most of the risk to the intensive animal industries [Hardefeldt et al., 2018b].

CONCLUSION

This study reveals that veterinary students at the University of Novi Sad have a relatively good understanding of AMR and AMS. However, more education on the appropriate use of antimicrobials is needed, as well as delivering adequate educational interventions in undergraduate program of veterinary medicine. This is the first study of this type conducted in Serbia.

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SENSORY EVALUATION OF DRY PERSIMMONS OF THE 'TIPO' (*DIOSPYROS KAKI* L. F.) FRUIT VARIETY

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Summary: *In the conducted study the persimmons of the 'Tipo' fruit variety were dried. Before they were dried, the fruit was either softened or not, cut to various thicknesses and shapes, and peeled or not. In January 2019 a hedonic sensory evaluation was performed, where evaluators evaluated the dried persimmons regarding the appearance (shape and colour) and taste (sugar/acid ratio, texture, tartness, presence of the fruit peel (only one test group), and general impression). The results indicated that dried unfertilized persimmons, with the peel cut into 3 mm thick slices, were given the highest score regarding their attractiveness. Regarding the taste parameters, the unfertilized peeled persimmon slices scored the highest, which is also the case in the category of overall impression. The average evaluation of all parameters indicates that the unfertilized persimmons got a higher evaluation than the fertilized persimmon. The evaluators assessed that the presence of the peel is not disruptive.*

Key words: *Persimmon / Drying Fruits / Hedonic Sensory Evaluation*

INTRODUCTION

Persimmon is a fruit that originates in China and grows in subtropical and temperate zone. In the last year the interest in persimmon has strongly risen. Consumers are interested because of its high nutrient levels and healing properties (Ambrožič Turk et al. 2009). Persimmon cultivars are classified into four types depending on the nature of astringency loss in the fruit. The 'Tipo' variety is the most used sort in Slovenia. It is a part of the "pollination variant non-astringent" group. Fruit is tart, if it is not fertilized, and they need to be softened to be consumable. If the fruit is fertilized, it can be consumed right at harvesting while it's still hard.

Drying is one of the ways to store fruit that is not permanently sustainable. At the same time it is the oldest, most natural and rational way of conserving foodstuff (Miklošič 1998). Conserving fruit with drying is based on deprivation of water from the regular 80-90% to 15-20%. That thwarts the basic criteria for microorganisms to develop and grow. The deprivation of water extends fruit's sustainability (Škof 2003).

The aim of an experiment made by Chung et al. (2017) was to figure out how the elimination of astringency before drying affects the colour, texture, tannin content and the sensory evaluation of dry persimmon. Evaluators graded the likeness of colour, aroma, texture, sugar content and general acceptability of dry persimmon. Dry persimmon that were exposed to ethanol received

the highest grades on all graded parameters. Exposing persimmon fruit to different factors before drying affects the quality of characteristics of dry persimmon.

Karakasova et al. (2015) used fresh persimmon and persimmon that was frozen for 24 hours for an experiment. Fresh and thawed persimmon was cut into pieces. The pieces of fresh and thawed persimmon were split into three groups and submerged into one of the solutions: citric acid, potassium metabisulfite ($K_2S_2O_5$) and ascorbic acid. With sensory grading they compared the dry persimmon in colour, taste and smell. The thawed persimmon had the best taste because there was no tartness. The best colour grades received fresh and thawed persimmon that was submerged into Potassium bisulphate, which keeps the bright colour.

The aim of this paper is to find out how dried persimmon of 'Tipo' variety, that differentiates by shape, thickness and pollination, affects the evaluators's grades on properties such as: shape, colour, ratio between sugar and acid, texture, tartness and general grade.

MATERIAL AND METHODS

Materials

We used persimmon of the 'Tipo' variety, that was grown in 2018 at farm Marko, that is situated in settlement Metava in municipality Maribor. Persimmon was harvested in two orchards. One contained only trees of the 'Tipo' variety and the other contained another sort 'Čokolatino'. Fruits were harvested in full ripeness (second half of November). They were stored in a cold storage room at 4 °C.

Preparation of dried persimmon

We used unfertilized persimmon in the first five treatments, fertilized fruit in the sixth, and fertilized and extra softened fruit. Fruit was left for five day to become soft at room temperature of 22 °C. Persimmon was washed and peeled (except in one treatment). Then it was sliced into three different thicknesses (3 mm, 7mm, 10mm) with the help of a meat slicer. In one treatment persimmon was cut into wedges. We weighed the initial mass of cut up persimmon and then dried them in a ventilation dryer at 40 °C until the persimmon weighed 20% of the initial mass. Dried fruit was stored in plastic hermetically sealed contained in a cold environment.

Sensory evaluation

65 evaluators participated in the evaluation. Each sample of dried persimmon was on its own plastic plate, marked with a three-digit code. Picture 1 shows some examples of persimmon, marked with codes. The samples were placed in a random order. Each evaluator started the evaluation with a different sample. The grade of attractiveness was passed in two parts. Attractiveness of the appearance (shape and colour), was evaluated on the same pieces of dried persimmon. Then each of the evaluators continued the evaluation of taste on his own spot (sugar/acid ratio, texture, tartness, presence of the skin – in one treatment, and general

impression. The grade of each of the parameters was marked with the letter I on a non-structured 10 cm scale with marked ends. The left end was marked with ‘does not please me’ (0 cm), the right end with ‘pleases me very much’ (10 cm). Evaluators had an option to add a note at each treatment.

Statistical analysis

The gathered results were input and sorted in Microsoft Office Excel 2007. The results were then statistically processed in the programme Statgraphics Centurion XV (Statgraphic 2005). To determine the statistically significant influence of the evaluator and treatment, we used Multifactor ANOVA. The results are shows as mean- values of grades of attractiveness with standard error of a mean ($\pm$ SEM). Differences between mean values were determined with a test of multiple comparisons (Duncan, $\alpha = 0,05$).



Photo 1: Seven treatments of dried persimmon, marked with three-digit codes, that differentiate in fertilization, thickness, the way of cutting and the presence of the skin. 829 – unfertilized, peeled, 7 mm thick slice; 397 – unfertilized, peeled, 3mm thick slice; 520 – unfertilized, peeled wedge; 735 – unfertilized, unpeeled, 7 mm thick slice; 461 – unfertilized, peeled, 10mm thick slice; 253 – fertilized, peeled, 7 mm thick slice; 682 – softened, fertilized, peeled, 7 mm thick slice.

RESULTS AND DISCUSSION

Subjective evaluator grade represents an important source of variation ($P \leq 0,001$), the reason is the differences in perception of each of the properties. Grades from 0 to 10 present the centimeters, that evaluators marked on the 10 cm grading line. Grades of attractiveness of colour and shape are shows in table 1. There is a statistic difference in treatments 1 and 2 in comparison with the others regarding the attractiveness of shape. Treatment 1 and 2 represent unfertilized, peeled persimmons. Evaluators gave the highest average grade on attractiveness of colour to

samples from treatment 1 (7,8). Statistically similar to treatment 1 are treatments 2 and 4. In all three treatments were unfertilized persimmons. Unfertilized persimmon has brighter yellow orange colour of the meat.

Table 1: Grades of attractiveness of appearance

Source of variation / Sensory property	Shape	Colour
Evaluator	3,19*** ¹⁾	2,29***
Treatment	51,43***	45,57***
	Mean value $\pm$ SEM	
Treatment ²⁾		
1	8,1 $\pm$ 0,19a ³⁾	7,8 $\pm$ 0,23a
2	7,6 $\pm$ 0,23ab	7,7 $\pm$ 0,24a
3	6,5 $\pm$ 0,25c	6,0 $\pm$ 0,26b
4	7,1 $\pm$ 0,22b	7,4 $\pm$ 0,21a
5	4,1 $\pm$ 0,32d	5,1 $\pm$ 0,30c
6	4,3 $\pm$ 0,26d	3,9 $\pm$ 0,30d
7	4,8 $\pm$ 0,33d	4,1 $\pm$ 0,34d

¹⁾ ***-statistically significant variation at $P \leq 0,001$.

²⁾ 1 – unfertilized, peeled, 3 mm thick slice; 2 – unfertilized, peeled, 7 mm thick slice; 3 - unfertilized, peeled, 10 mm thick slice; 4 - unfertilized, unpeeled, 7 mm thick slice; 5 - unfertilized, peeled wedge; 6 – fertilized, peeled, 7 mm thick slice; 7 - softened, fertilized, peeled, 7 mm thick slice.

³⁾ Mean values ($\pm$ SEM), marked with different letters, have statistically significant differences (Duncan test, $\alpha = 0,05$).

The highest average grade (7,3), at parameter sugar/acid ratio, received treatments 3 and 5. Treatment 3 represents samples that were cut into slices with a thickness of 10 mm, treatment 5 represents samples that were cut into wedges. Both of these treatments had the highest thickness, which is why the samples were dried for a longer period. Statistically significant similar grades of attractiveness got the samples of treatments 2 and 4, which were made from unfertilized fruit. The highest average score for texture got a slice of unfertilized persimmon with a thickness of 7 mm. Treatment 7 received statistically the worst average grades (4,8). Treatment 7 contained samples that were fertilized and extra softened fruit. Table 2 shows that the highest average grade of attractiveness based on astringency received persimmon that was cut into wedges (7,2). Statistically similar to treatment 5, are the sample from treatment 2,3,4 and 6. Evaluators liked the astringency at all 5 treatments that contained unfertilized persimmon. These fruits were inedible before drying because of the high value of tannin, that prevents eating unless the fruit is softened. Presence of the skin of persimmon was asked about specifically at samples from treatment 4. The average grade for the presence of the skin was 5,3. Evaluators wrote in the additional noted that as long as the skin is thin enough, it completes the shape of dried persimmon.

Table 2: Grades of attractiveness of taste

Source of variation / Sensory property	Sugar/acid	Texture	Astringency
Evaluator	3,26*** ¹⁾	3,23***	5,42***
Treatment	11,86***	11,23***	3,85***
	Mean value $\pm$ SEM		
Treatment ²⁾			
1	6,3 $\pm$ 0,28b ³⁾	6,8 $\pm$ 0,29ab	6,3 $\pm$ 0,31bc
2	6,9 $\pm$ 0,24ab	7,3 $\pm$ 0,23a	6,7 $\pm$ 0,29ab
3	7,3 $\pm$ 0,22a	6,8 $\pm$ 0,32ab	7,1 $\pm$ 0,31a
4	7,1 $\pm$ 0,24a	6,1 $\pm$ 0,30bc	6,7 $\pm$ 0,31ab
5	7,3 $\pm$ 0,28a	7,0 $\pm$ 0,30a	7,2 $\pm$ 0,31a
6	5,6 $\pm$ 0,33c	5,7 $\pm$ 0,31c	6,4 $\pm$ 0,30abc
7	5,3 $\pm$ 0,30c	4,8 $\pm$ 0,33d	5,8 $\pm$ 0,33c

¹⁾ ***-statistically significant variation at $P \leq 0,001$.

²⁾ 1 – unfertilized, peeled, 3 mm thick slice; 2 – unfertilized, peeled, 7 mm thick slice; 3 - unfertilized, peeled, 10 mm thick slice; 4 - unfertilized, unpeeled, 7 mm thick slice; 5 - unfertilized, peeled wedge; 6 – fertilized, peeled, 7 mm thick slice; 7 - softened, fertilized, peeled, 7 mm thick slice.

³⁾ Mean values ($\pm$ SEM), marked with different letters, have statistically significant differences (Duncan test, $\alpha = 0,05$).

General impression represents the wholesome eating quality. Statistically significant, the best graded treatments were those that contained the unfertilized persimmon. The worst graded persimmon was fertilized and extra softened persimmon.

Table 3: Grades of attractiveness of general impression

Source of variation / Sensory property	General impression
Evaluator	2,97*** ¹⁾
Treatment	11,39***
	Mean value $\pm$ SEM
Treatment ²⁾	
1	6,6 $\pm$ 0,29a ³⁾
2	6,9 $\pm$ 0,27a
3	6,8 $\pm$ 0,27a
4	6,5 $\pm$ 0,28a
5	7,0 $\pm$ 0,28a
6	5,5 $\pm$ 0,31b
7	4,7 $\pm$ 0,32c

¹⁾ ***-statistically significant variation at $P \leq 0,001$.

²⁾ 1 – unfertilized, peeled, 3 mm thick slice; 2 – unfertilized, peeled, 7 mm thick slice; 3 - unfertilized, peeled, 10 mm thick slice; 4 - unfertilized, unpeeled, 7 mm thick slice; 5 - unfertilized, peeled wedge; 6 – fertilized, peeled, 7 mm thick slice; 7 - softened, fertilized, peeled, 7 mm thick slice.

³⁾ Mean values ($\pm$ SEM), marked with different letters, have statistically significant differences (Duncan test, $\alpha = 0,05$).

CONCLUSION

From the gathered results of sensory evaluation of dried persimmon of 'Tipo' variety, we have discovered that evaluation prefer the appearance of unfertilized persimmon that is cut into slices. Unfertilized slices are joined by the unfertilized wedge, that was the highest graded by the attractiveness of taste. The wedge was also the best graded in the general impression part of the evaluation.

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CONSUMER RECOGNITION AND UNDERSTANDING OF PREPACKED FOOD LABELS AMONG STUDENTS AND STAFF AT THE FACULTY OF AGRICULTURE AND LIFE SCIENCES

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Summary: Regulation (EU) No 1169/2011 specifies which information must be present on products. Besides the mandatory labels, we can also find additional labels, which describe a higher quality of products or special treatment of products during processing. The amount of information on the products is immense, which allows consumers to choose the one that suits them the most, but they often do not understand or recognize them. The aim of this thesis was to find out if employees and students of Faculty of Agriculture and Life Sciences know which labels must be placed on products, if they know any additional labels and how often they buy products with them. A statistical difference appeared between students and employees in recognizing and understanding the EU organic logo and in the nutrition labelling (for unsaturated fats), while the other marks were equally recognized and understood. Results also showed that gender partially effects the understanding of labels. Students from the master's program Food safety in the Agri-food Chain did not statistically deviate from other students. Employees and students have similarly decided to purchase products with additional labels.

Key words: Labelling / Prepacked Foodstuff / Surveying / Consumers

INTRODUCTION

There are more and more labels on prepacked foodstuffs. Regulation (EU) 1169/2011 (2011) determines which information is obligatorily on prepacked foodstuffs. On prepacked foodstuffs we can also find voluntary labels, regulated with other laws (i.e. nutritional and health claims) and labels that show off a higher quality or special treatment of the foodstuffs (i.e. EU quality schemes, no GMO labels).

On prepacked foodstuffs can be a lot of information, that give consumers possibility to compare different products and to choose the right product for themselves. Although, consumers often do not know the labels or their meaning.

The aim of this paper was to find out if students of different studying programs and employees of Faculty of Agriculture and Life Sciences know, which labels have to be on foodstuffs and if they know voluntary labels, their meanings and how often do they buy foodstuffs with such labels.

For the research we did a survey and analyzed answers about recognition and understanding of prepacked food labels. The survey was made among students and employees at the Faculty of Agriculture and Life Sciences. We analyzed the answers in connection with shopping habits and socio-demographic differences of the consumers surveyed.

Hypotheses

H1: Faculty of Agriculture and Life Sciences employees are more familiar with and understand the labeling more on prepackaged foods than students.

H2: Students in the MA program Food Safety in the Food Chain are more familiar with and understand the labels more on pre-packaged foods than students in other degree programs.

H3: Knowledge and understanding of information on prepackaged foods is statistically significantly influenced by the gender and age of the consumer. We believe that among the respondents, better results will be achieved by female consumers, among students in the last year of undergraduate or postgraduate studies.

H4: Employees are more likely to buy foods with voluntary labels than students.

MATERIAL AND METHODS

The survey was conducted in the summer of 2019 among students and employees of the Faculty of Agriculture and Life Sciences of the University of Maribor.

Survey questionnaire was created after a questionnaire that was used by student Kozmos (2018). It was made with the help from a web tool called 1ka.si. The questionnaire was solved online in most cases, except by 6 respondents.

The questionnaire had 36 questions, that were divided in 3 sections. In first section were 16 questions about consumers knowledge and understanding of labels on prepacked foodstuffs. With pictures of labels we tested consumers recognition and knowledge about their meaning. In addition, we were interested in their knowledge of labels that have to be on prepacked foodstuffs (i.e. allergens, ingredients, date of minimum durability).

In the second section we asked the consumers 4 questions about shopping and eating habits, how much money do they spend for foodstuffs in an average month and if they avoid any of the foodstuff.

In the third section we asked seven demographic questions (gender, region of habitation, number of people in their household, net monthly income of the respondent, employment status, highest degree of education achieved, ...).

We exported the information from the web tool into program Microsoft Office Excel, where we coded it for further processing in program SPSS 25.0 for program environment Windows.

We did statistic processing of the information in two parts. First, we described the sample from socio-demographic point of view and showed the results in frequency and shares. In second part we tested the hypotheses with Chi-square test of independence (χ^2). Because of immense amount of information, we showed only results of Chi-square tests of independence, for which we've noticed significant difference at significance level $p < 0,05$. In first three hypotheses (H1 — H3) we've tested a group of questions which refer to recognition and understanding of labels on prepacked food (questions 1, 6, 8, 9, 10 and 15 of the survey). In H4 we firstly tested if it was Gauss distribution with Kilmogorov-Smirnov's test. Results of this test showed, that it wasn't a Gauss distribution, that is why we've decided to use non-parametric Mann-Whitney's U test.

RESULTS AND DISCUSSION

With Chi-square test of independence, we tested hypotheses H1, H2 and H3. We included questions listed below (1, 6, 8, 9, 10 and 15) in statistical analysis. We divided H3 in two sections. First, we tested if gender of respondents affects the recognition and understanding of labels on prepacked food, then we tested if gender and level of education affects the recognition and understanding of labels on prepacked food.

Q1: Have you ever seen the labels below on any foodstuffs? If you have already seen the label and you know what it means, write the meaning in the third column.

H1: There is a significant difference only in understanding of European label for organic foodstuffs ($p = 0,018$; $p < 0,05$). Students (46,9 %) understand the label better than employees (40,0 %). There is no difference in understanding other labels between students and employees.

H2: There is no significant difference between students of MA program Food Safety in the Food Chain and students of other degree programs.

H3: The only significant difference between female respondents and male respondents was in label for Halal foodstuffs ($p = 0,29$; $p < 0,05$). Men (38,5 %) knew the label Halal better than women (28,0 %). There was no significant difference in other labels. There is also no significant difference between younger and older students.

Q6: »In table below is information that appears on labels of prepacked foodstuffs. Mark the ones you think are obligatory information«.

From 15 written answers, 10 of them were correct.

H1: There is no significant difference between students and employees.

H2: There is no significant difference between students of MA program Food Safety in the Food Chain and students of other degree programs.

H3: More female respondents (48,0 %) thought that the daily intake of energy and nutrients should be written on packaging than male respondents (33,0 %) ($p = 0,027$; $p < 0,05$). There is no significant difference between younger and older students.

Q8: “Which substances must be included in nutrition declaration?”

There were 17 substances, the correct ones were salt, proteins, fats, carbon hydrates, energy, sugars and saturated fats.

H1: There was a significant difference between employees and students in unsaturated fats. Because the labelling of unsaturated fats is not obligatory, students answered better than employees in this segment.

H2: There is no significant difference between students of MA program Food Safety in the Food Chain and students of other degree programs.

H3: Female respondents answered better than male respondents that information about yeast and additives doesn't have to be on food labels.

Q9: “On gummy candy there is a date of minimum durability 30th September 2019. Can we consume it a week after it expires?”

Correct answer is “Yes, after expired date of minimum durability this foodstuff remains same quality.”

H1: There is no significant difference between students and employees.

H2: There is no significant difference between students of MA program Food Safety in the Food Chain and students of other degree programs.

H3: There is no significant difference between female and male respondents nor between older and younger students.

Q10: “On a yoghurt there is “use by” dated 30th September 2019. Can we consume it a week after it expires?”

Correct answer is: “No, this foodstuff, even if stored correctly, is not safe to consume after the “use by” date expires. “

H1: There is no significant difference between students and employees.

H2: There is no significant difference between students of MA program Food Safety in the Food Chain and students of other degree programs.

H3: There is no significant difference between female and male respondents nor between older and younger students.

Q15: Mark the foodstuffs that have to be labelled as “GMO foodstuffs” (that they contain genetically modified organisms) or that they were produced with GMO.

Correct answer is: “Plant foodstuffs containing more than 0,9 % of GMO”

H1: There is no significant difference between students and employees.

H2: There is no significant difference between students of MA program Food Safety in the Food Chain and students of other degree programs.

H3: There is no significant difference between female and male respondents nor between older and younger students.

In H4 we first tested if it was Gauss distribution with Kilmogorov-Smirnov's test. Results of this test showed that it wasn't a Gauss distribution ($p > 0,01$). That is why we've decided to use non-parametric Mann-Whitney's U test for validation of differences between choosing foodstuffs with optional labels among employees and students. There was no significant difference ($p > 0,05$) in any of the given labels, which means that students and employees at Faculty for Agriculture and Life Sciences choose foodstuffs with optional labels at almost same frequency.

CONCLUSION

In a survey, conducted among students and employees at Faculty for Agriculture and Life Sciences, we wanted to find out how well students and employees recognize and understand the meaning of obligatory labels, labels of higher quality and what are their shopping habits. 114 respondents participated in survey, 74 of them were students and 40 were employees. 39 of respondents were male and 75 were female.

Statistical analysis with non-parametric tests showed that:

- H1 can be completely rejected, because the statistically significant differences between students and employees at Faculty of Agriculture and Life Sciences in knowing and understanding the labels on prepackaged foodstuffs do not exist, except in the case of European label for organic foodstuffs and in labelling of unsaturated fats, where students answered better than employees
- H2 can be completely rejected, because the statistically significant differences between students of MA program Food Safety in the Food Chain and students of other degree programs at Faculty of Agriculture and Life Sciences in knowing and understanding the labels on prepackaged foodstuffs do not exist.
- H3 can be completely rejected; we divided it in:
 - o H3a, where we tested the effects of gender on recognizing and understanding labels on prepacked foodstuffs. We rejected it, because male respondents answered more correctly than female respondents about the label Halal and they knew more about obligatory information on prepackaged food (a lot of the female respondents thought that dairy intake of energy and nutrients is obligatory information). Female respondents answered better only at the label of nutritional value (yeast and additives were correctly put into the non-obligatory information).
 - o H3b, which can be completely rejected, because the statistically significant differences between older female students and younger students at Faculty of

Agriculture and Life Sciences in knowing and understanding the labels on prepackaged foodstuffs does not exist.

- H4 can be completely rejected, because the statistically significant differences between students and employees at Faculty of Agriculture and Life Sciences in frequency of choosing prepackaged foodstuffs with optional labels does not exist

We can conclude that students and employees of Faculty of Agriculture and Life Sciences recognize and understand labels on prepackaged foodstuffs very well. Students likely know this because of their education at FKBPV. Another reason for such a good knowledge about labels is that more than a third of respondents (39%) regularly checks the labels on packaging while shopping.

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BIRDS FROM A MINI ZOO IN SERBIA AS A SOURCE OF ANTIBIOTIC-RESISTANT ENTERIC BACTERIA

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Summary: *Antimicrobial resistance is one of the major threats to public health affecting humans, animals, and the environment. Of particular concern is the potential transmission of multidrug-resistant zoonotic pathogens from zoo animals to humans. The aim of our research was to determine the antibiotic resistance patterns in enteric bacteria isolated from zoo birds. This study included 12 bacterial strains isolated from 12 birds from a mini zoo in Serbia. For the antimicrobial susceptibility testing of the bacterial isolates the disc diffusion method was used according EUCAST guidelines. Highest resistance was observed to ampicillin (50%), while the lowest resistance was observed to cefuroxime, ciprofloxacin, ceftriaxone and gentamicin (8.33%), with two multi drug resistant strains. To our knowledge, this study was conducted as the first study on this topic in Serbia. The occurrence of antimicrobial resistance is an interesting fact given that the birds were never treated with antibiotics. Cohabitation among different bird species and also contact between wildlife and people who often visit the zoo indicate that contact between them can be an important path of transferring genes of resistance. Considering the small number of samples included in this research further studies are needed with a larger number of samples and other classes of antibiotics in order to gain more clear picture on the abundance and prevalence of antibiotic-resistant enteric bacteria in birds in Serbia.*

Key words: *enteric bacteria, birds, antibiotic resistance, Serbia*

INTRODUCTION

Antimicrobial resistance (AMR) is one of the major threats to public health affecting humans, animals, and the environment, thus the intervention requires a one health approach [World Health Organization, 2014]. Problems associated with the development and spread of antibiotic resistance in clinical practice have been increasing since the early 1960s and are currently viewed as major concerns of the public health on a global level [Levy et al, 2004].

The bacteria of the normal flora of the gut, such as f. *Enterobacteriaceae*, can easily acquire and transfer resistance genes. These commensal bacteria, which represent a reservoir of resistance genes for pathogenic bacteria, can thus be used as indicators of changes in antimicrobial resistance [Caprioli et al., 2000]. Extended-spectrum β -lactamases are responsible for resistance to most β -lactam antibiotics (e.g. penicillins and cephalosporins) and are

predominantly found among *Enterobacteriaceae* leading to the spread of resistance genes and to serious infections [Zhang et al., 2015].

Of particular concern is the potential transmission of multidrug-resistant zoonotic pathogens from zoo animals to humans. One report identified more than 25 human infectious disease outbreaks over a 10-year period (1990 to 2000) as being associated with visits to animal exhibits [Bender et al., 2004].

As a little is known about antimicrobial-resistant bacteria in zoo birds and to our knowledge there is no published data on this topic in Serbia, the aim of our research was to determine the antibiotic resistance patterns in enteric bacteria isolated from zoo birds.

MATERIAL AND METHODS

For the purpose of this study stool samples from 12 birds, 11 different species of birds respectively (Table 1.), were collected at the mini zoo "Miki" in Kolut, Serbia. All the samples were collected in sterile plastic dishes and within four hours were transported to the laboratory of the Department of veterinary medicine, Faculty of Agriculture, University of Novi Sad for further processing. All of the samples were inoculated to Endo agar (Biolife Italiana, Milan, Italy) and left for incubation at 37° C for 24 hours in air. After cultivation, the identification of bacteria was based on biochemical testing that included test of indole production, motility and H₂S production on SIM medium (HiMedia, Mumbai, India), citrate utilisation test on Simmon's citrate agar (HiMedia, Mumbai, India) and double sugar fermentation test and H₂S production on Kligler iron agar (Biolife Italiana, Milan, Italy). All mediums were prepared according to the manufacturer's instructions.

Table 1. Sample numbers, bird species and species of isolated bacteria.

Sample number	Species	Bird species	Species of isolated bacteria
1	<i>Pipile cumanensis</i>	Blue-throated piping guan	<i>Proteus spp.</i>
2	<i>Corvus albus</i>	Pied crow	<i>Escherichia coli</i>
3	<i>Guttera pucherani</i>	Crested guineafowl	<i>Escherichia coli</i>
4	<i>Bycanistes brevis</i>	Silver-cheeked hornbill	<i>Escherichia coli</i>
5	<i>Crax rubra</i>	Great currasow	<i>Escherichia coli</i>
6	<i>Psittacus erithacus</i>	Grey parrot	<i>Klebsiella spp.</i>
7	<i>Anas platyrhynchos dom.</i>	Domestic duck	<i>Escherichia coli</i>
8	<i>Balearica regulorum</i>	Grey crowned crane	<i>Escherichia coli</i>
9	<i>Phoenicopterus roseus</i>	Greater flamingo	<i>Enterobacter spp.</i>
10	<i>Phoenicopterus roseus</i>	Greater flamingo	<i>Escherichia coli</i>
11	<i>Anser canagicus</i>	Emperor goose	<i>Escherichia coli</i>
12	<i>Branta ruficollis</i>	Red breasted goose	<i>Escherichia coli</i>

For the antimicrobial susceptibility testing of the bacterial isolates the disc diffusion method was used according to the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines [EUCAST, 2017]. Müller-Hinton agar (HiMedia, Mumbai, India) was prepared according to the manufacturer's instructions and stored in Petri dishes. The inoculum was prepared using the direct colony suspension method to make a suspension of the organism in saline to the density of the McFarland 0.5 turbidity standard, approximately corresponding to $1-2 \times 10^8$ CFU/ml for *Escherichia coli*. Prior to inoculation, the agar plates were kept at room temperature. The inoculum suspension was used within 15 minutes of the preparation by dipping a sterile cotton swab into the suspension. To avoid over-inoculation of Gram negative bacteria, the excess of fluid was removed by pressing and turning the swab against the inside of the tube. The inoculum was spread evenly over the entire agar surface ensuring that there are no gaps between the streaks. Antimicrobial disks (Bioanalyse, Ankara, Turkey) were applied within 15 minutes of inoculation, after which the agar plates were incubated for 16-20 hours at $35 \pm 1^\circ\text{C}$ in air. Antimicrobial susceptibility of the bacterial isolates was tested for the following classes of antibiotics: penicillins (ampicillin 10 µg, amoxicillin with clavulanic acid 20+10 µg), cephalosporins (cefuroxime 30 µg, ceftriaxone 30 µg), macrolides (gentamicin 10 µg), sulfonamides (sulfamethoxazole with trimethoprim 1,25+23,75 µg) and quinolones (ciprofloxacin 5 µg). After the incubation the inhibition zones of bacterial growth around the applied discs were measured and interpreted according to the EUCAST guidelines [EUCAST, 2017].

RESULTS AND DISCUSSION

AMR is one of the most serious emerging health threats, and is studied extensively in both humans and animals because of its nature and extent [Sala et al., 2016]. To our knowledge, this study was conducted as the first study on this topic in Serbia with the aim to determine antibiotic resistance patterns in enteric bacteria isolated from zoo birds.

In this study, antimicrobial resistance was noted for each tested antibiotic, but four isolates (33.33%) from samples no. 1, 3, 5 and 10 showed susceptibility to all used antibiotics.

Highest resistance was observed to ampicillin (50%), followed by amoxicillin with addition of clavulanic acid (33.33%), sulfamethoxazole with trimethoprim (16.66%), with the lowest resistance to cefuroxime, ciprofloxacin, ceftriaxone and gentamicin (8.33%).

Multidrug resistance (MDR), defined as resistance against at least one antibiotic in at least three classes of antibiotics, was found in two (16.66%) isolates of *E. coli*, samples 7 and 11 respectively.

Extended-spectrum β -lactamases (ESBL) were noted in two samples (16.66%), i.e. sample 7 (resistant to ampicillin and cefuroxime) and 9 (resistant to ampicillin, amoxicillin with addition of clavulanic acid and ceftriaxone) both resistant to the representatives of penicillin and cephalosporin class of antibiotics.

The results of the antibiogram are shown in Table 2.

Table 2. Patterns of antibiotic resistance of bacterial isolates

Sample number	AMP	AMC	CXM	CRO	CN	SXT	CIP
1	S	S	S	S	S	S	S
2	R	R	S	S	S	R	S
3	S	S	S	S	S	S	S
4	R	S	S	S	S	S	S
5	S	S	S	S	S	S	S
6	S	R	S	S	S	S	S
7	R	S	R	S	I	S	S
8	R	S	S	S	S	S	S
9	R	R	S	I	S	S	S
10	S	S	S	S	S	S	S
11	R	R	S	S	S	R	R
12	S	S	S	S	S	S	S

Legend: AMP – ampicillin, AMC – amoxicillin with clavulanic acid, CXM – cefuroxime, CRO – ceftriaxone, CN – gentamicin, SXT – sulfamethoxazole with trimethoprim, CIP – ciprofloxacin, S – sensitive, R – resistant, I – intermediate

Data on this topic in the world in general is scarce and considering the differences in geographic areas and differences of bird species included in studies, it is difficult to compare prevalence rates of antibiotic resistance obtained in our study with the results of other authors.

Four *E. coli* isolates (33.33%) used in this study showed sensitivity to all antibiotics, whereas 50% of isolates showed resistance to ampicillin. In a study from Portugal, 97.22% *E. coli* isolates showed resistance to one or more antibiotics. More than 60% of the isolates were resistant to ampicillin, similar to the results of our study. [Radhouani et al, 2012].

In our study, ESBL was noted in two samples (16.66%) which is higher prevalence rate than 1.7% reported by Zurfluch et al. in the study conducted on common wild bird species in Switzerland. [Zurfluch et al., 2019]. In contrast to the results of our study, no ESBL producing *E. coli* was isolated from wild birds in a veterinary hospital in São Paulo, Brazil by Borges et al., however multidrug resistance was associated with 47.4% of these isolates, in contrast to the prevalence of 16.66% in our study. Interestingly, the same study from Brazil a higher prevalence of resistance to ciprofloxacin (36.8%) and gentamicin (21%) was observed but also lower to ampicillin (21%) and amoxicillin with clavulanic acid (15.8%) which is also in contrast to the prevalence obtained in our research [Borges et al., 2017]. In another study on *E. coli* in captive birds in São Paulo by Pontes et al., the highest resistances were observed with amoxicillin with clavulanic acid and ampicillin, and MDR was verified in 59% of the isolates, which is in agreement to our results [Pontes et al., 2018].

The second highest pattern of resistance was observed to amoxicillin with clavulanic acid with prevalence of 33.33%, followed by 16.66% to sulfamethoxazole with trimethoprim, whereas in study from Istanbul, Turkey a higher percentage of resistance to sulfamethoxazole with

trimethoprim (46.66%) than amoxicillin with clavulanic acid (13%) was observed [Sigirci et al., 2019].

In the present study we have found one isolate (8.33%) resistant to ciprofloxacin, whereas in a study from Brazil all of the tested *E.coli* strains showed susceptibility to this antibiotic [Machado et al, 2018].

CONCLUSION

Based on the results obtained, antimicrobial resistance was noted in enteric bacteria isolated from birds in mini zoo "Miki" in Kolut, Serbia. In particular, the greatest resistance was observed to ampicillin. The occurrence of AMR is an interesting fact given that the birds were never treated with antibiotics. Cohabitation among different bird species and also contact between wildlife and people who often visit the zoo indicate that contact between them can be a important path of transferring genes of resistance. Considering the small number of samples included in this research further studies are needed with a larger number of samples and other classes of antibiotics in order to gain more clear picture on the abundance and prevalence of antibiotic-resistant enteric bacteria in birds in Serbia.

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THE NEW TECHNIQUE OF HERNIORRHAPHY USED IN RECURRENT PERITONEOPERICARDIAL HERNIA IN A TWO-YEARS-OLD RHODESIAN RIDGEBACK – A CASE REPORT

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Summary: *Peritoneopericardial hernia is a malformation which allows the protrusion of abdominal organs into the pericardial sac. Several methods of herniorrhaphy had been described in human and veterinary medicine, such as the usage of prosthetic patches, allografts and autologous grafts. The purpose of this study was to describe a new technique of herniorrhaphy, which combines two different materials for defect closure (nonautologous-polypropylene mesh and autologous-pericardial flap) in a case of recurrent peritoneopericardial hernia in a dog. A two-year-old, intact female Rhodesian Ridgeback was presented with an eight week history of respiratory problems, occasional coughing, vomiting, lethargy, body mass loss and icteric mucous membrane. The reason for using this technique was tension at hernial ring, which could have caused recurrence of the peritoneopericardial hernia. Polypropylene mesh was used in order to reinforce the herniorrhaphy and an autologous flap served as a barrier against mesh exposure to the epicardial surface. In our opinion, this method is safe with no complications observed, so therefore, it could be a suitable method for peritoneopericardial reherniation repair.*

Key words: *peritoneopericardial hernia, herniorrhaphy, pericardial flap, mesh, repair*

INTRODUCTION

Peritoneopericardial diaphragmatic hernia (PPDH) is the most common congenital defect involving the pericardium of dogs and cats (ROSENSTEIN et al., 2001; EVANS and BIERLY, 1980; BELLAH et al., 1989). Peritoneopericardial diaphragmatic hernia has also been reported in rabbits, in a donkey, and a calf (KHEIRANDISHI et al., 2014). In this anomaly, abdominal contents are herniated into the pericardial sac because of direct communication between the peritoneal and the pericardial cavities. An embryonic development defect of the dorsolateral septum transversum in the so-called sternocostal triangle structure can cause this developmental abnormality. Peritoneopericardial diaphragmatic hernia occurs in both sexes, and an autosomal recessive mode of inheritance has been postulated. The defect consistently involves the left dorsolateral and central portions of the diaphragm, with herniation of abdominal organs into the thoracic cavity. This malformation allows the passage of abdominal contents through the

diaphragmatic defect into the pericardial sac. Passage of the liver, spleen, stomach, or intestines into the pericardial sac may result in cardiac tamponade, compromised function of the respiratory tract, gastrointestinal tract obstruction, gallbladder torsion, and splenic or hepatic entrapment (SCHULMAN et al., 1985; REIMER et al., 2004). Persistent communication between the peritoneal and pericardial cavities allows the abdominal contents to herniate into the pericardial sac. Peritoneopericardial hernia is frequently associated with supraumbilical hernia, and malformation or absence of the sternbrae and pectus excavatum. Other potential causes for PPDH include malformations and teratogen-induced factors (BERRY et al., 1990). The present case represents an, until now, undescribed method of surgical management of a large, recurrent peritoneopericardial hernia. A combination of autologous pericardial tissue and polypropylene mesh was used in the surgical repair of this complicated PPDH reherniation.



Fig.1. Dorsoventral projection of the thoracic and abdominal radiograph, revealing a contrastagent filled mass, protruding cranially through the diaphragm. The cardiac silhouette is enlarged. The lesion is consistent with a peritoneopericardial diaphragmatic hernia with herniation of the pyloric part of the stomach and the proximal duodenum (arrows).

MATERIALS AND METHODS

The patient was treated with an intravenous infusion of lactated Ringer's solution at 2 mL/kg/h, and esomeprazole (Nexium, Astra Zeneca) in a dose of 0.5 mg/kg i/v for three days. The dog was premedicated with midazolam (0.3 mg/kg i/m) (Dormicum, Roche) and fentanyl (3 µg/kg i/v) (Fentanyl, Janssen). Anaesthesia was induced with propofol (Propofol MCT; Fresenius) in dose of 4 mg/kg i/v, and maintained with sevoflurane (Sevorane, Abbot) in oxygen under controlled ventilation, and a constant rate infusion of fentanyl in dose of 0.3 µg/kg/min i/v was provided. Amoxicillin and clavulonic acid (Augmentin, Glaxo Smith Kline) were used before and after surgery, in a dose 20 mg/kg i/v. After preparation of the ventral part of the chest and abdomen, a skin incision and a ventral midline coeliotomy were performed from just caudal

to the xiphoid to the umbilicus. On abdominal exploration, a well-marginated 7x7 cm defect was identified in the ventral right muscular part and the central tendineous part of the diaphragm. All liver lobes were in the abdominal cavity. The pericardium was filled with 200 mL of serous fluid. A round-shaped part of the spleen, as well as the pyloric part of the stomach with the cranial part of duodenum, were found to be herniated through the diaphragmatic defect in the pericardium. The gall bladder was enlarged. The base of the planned pericardial flap was attached at the left side of the diaphragm laterally. A stay suture was used to hold the pericardium. A part of pericardium was cut (7×7 cm) and rotated across the diaphragmatic defect on the right side. The pericardial flap was taken carefully and the mechanical ventilation stopped a few times for 30 seconds each during the pericardial incision to prevent iatrogenic lung laceration. Polypropylene mesh (Surgipro, Covidien) of the same dimension was cut. A diaphragmatic herniorrhaphy was performed using a simple continuous pattern with nonabsorbable suture material (Dermalon 2-0, Covidien). The pericardial flap and the polypropylene mesh were sutured to the diaphragm. The polypropylene mesh was positioned on the abdominal diaphragmatic side and the pericardial flap on the thoracic diaphragmatic side. A prophylactic incisional gastropexy was performed (Maxon 2-0, Covidien). The abdomen was lavaged with warm sterile saline, and closed routinely. Fentanyl was used for postoperative analgesia in a dose of 0.05 µg/kg/min next 24 hours. A transdermal fentanyl patch (Durogesic patch, 75 µg/h, Janssen) was also used for postoperative analgesia. Meloxicam was administrated in a dose 0.1 mg/kg for the first three days i/v (Meloxidolor, Genera), followed by peroral syrup. The following day, the patient had improved ambulation and was willing to eat.

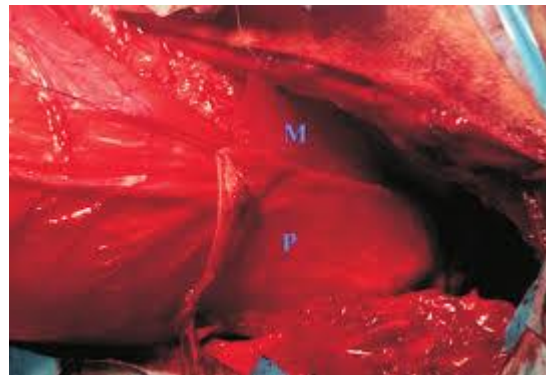


Fig.2. Surgical procedure of herniorrhaphy; positioning of the pericardial flap and polypropylene mesh. Part of the pericardium (adjacent to the hernial ring) was cut, with dimensions of 7×7 centimetres, and rotated across the defect at the right side (P). A polypropylene mesh of the same dimensions was cut (M). Cranial to the right.

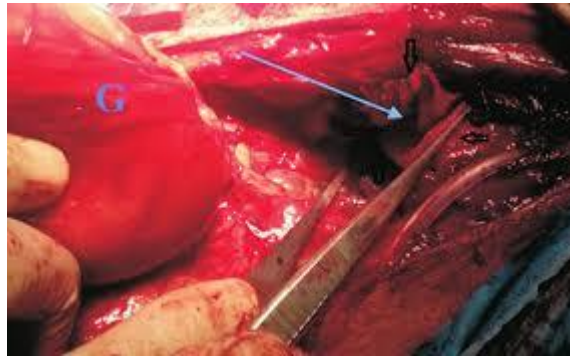


Fig.3. Surgical procedure of herniorraphy. The pericardial flap and polypropylene mesh were sutured to the diaphragm. The polypropylene mesh was positioned on the abdominal diaphragmatic side (long blue arrow) and the pericardial graft on the thoracic diaphragmatic side. Stomach (G). Diaphragmatic borders (black short arrows). Cranial to the right.

RESULTS AND DISCUSSION

Several methods of diaphragmatic herniorraphy had been described in veterinary medicine, such as usage of prosthetic patches (polytetrafluoroethylene, collagen coated vicryl mesh) (LALLY et al., 1996), and autologous flaps/grafts (fascia lata, flaps using the latissimus dorsi muscle, serratus muscle, internal oblique muscle and transversus abdominis muscle, pericardium) (WALLACE et al., 1992; SUZUKI et al., 2002; LIPTAK and BREBNER, 2006). In the present case, the tensile strength on the hernial ring was extensive, which could have caused disruption of the repaired hernia. Therefore, we decided to strengthen and bridge the repaired defect layers with polypropylene mesh, combined with pericardial flap. The pericardial flap is a strong fibrous tissue and its location close to the hernial ring, together with use of polypropylene mesh provided a suitable method of herniorrhaphy. Mortality may occur in up to 14% of dogs and cats undergoing surgery for PPDH (REIMER et al., 2004). In the present case, an autologous pericardial tissue and polypropylene mesh technique was used successfully as surgical treatment of a recurrent PPDH.

CONCLUSION

The diagnosis of the peritoneopericardial hernia is based on radiographic, ultrasonograph and surgical findings. In this case, the pyloric part of the stomach, the proximal part of the duodenum with the common bile duct and spleen were found in the peritoneopericardial hernial sac. The icteric mucous membranes, elevated ALT and hyperbilirubinemia observed in our case were the consequence of the kinked common bile duct, due to the abnormal position of the

pyloric part of the stomach and duodenum. Proximal gastrointestinal segment entrapment potentially caused nausea and vomiting due to delayed gastric emptying. The intrathoracic splenic position could have caused mesenteric tension, resulting in abdominal pain manifested by kyphosis. We did not observe any intra- or postoperative complications in the present case, and therefore this surgery could represent a good choice for PPDH repair, especially in a case of recurrence.

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SURGICAL THERAPY OF CLEFT PALATE IN DOGS

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Summary: *The palate in dogs is consisted by dorsal boundary of oral cavity. It is also separated in two parts. The hard palate part separates oral from nasal cavity. Cleft palate in dogs is congenital disorder. Certain disorder arises in fetal and embryonic period where we have incomplete closing of palatal process of maxilla and dorsal part of oral cavity. Unfortunately, this disorder is usually spread out through the whole length of hard palate, and it also overtakes the soft palate. With specific clinical signs we can easily diagnose this congenital problem. Clinical signs are very specific like inappropriate weaning or hardly swallowed food. The worst part is complication as aspiration pneumonia. The aim of this research is to present two cases of dogs with cleft palate. One of them was bull terrier (8 months old) and the other one was french bulldog (4 months old). The surgical procedure was implemented on Veterinary clinic for small animals, department of Veterinary medicine from Novi Sad.*

Key words: palate, congenital disorder, surgical procedure

INTRODUCTION

Congenital disorders of oral cavity have certainly usual appearance in dogs. The causes of congenital disorders could be from different sources. We can prove that endogenous and exogenous factors have very important influence in development of oral cavity in fetal period (Warzee, 2011). Cleft palate is considered as the most common congenital disorder. Except clefts, we can also have some other appearance such as macroglossy, microglossy, tight-lip syndrome and etc (Spasojević and Trailović, 2018). As for dogs, these disorders also have everyday appearance in other animal species. Nowadays, in domestication process of dogs, as home pets and in veterinary science developing we can easily access to exact type of healing. Palate in dogs is consisted by dorsal boundary of oral cavity which separates oral from nasal cavity (Gledić, 2012). In anatomic way it is separated on hard and soft palate. Hard palate is build by maxillar processus, while soft palate is structure of soft tissue and it has an important role in swallowing. From statistic datas, the appearance of cleft palate is between 25th and 28th day of fetal period with characteristic clinical signs (Evans, 1993). In addition to surgical procedures of healing, we can also apply some relatively new techniques which are not invasive as surgical (Ellion et.al., 1986). In prevention of disease it is also indicated to use mesenchyme pluripotent cells of iliac bone or we can use orthodontic braces (Mossey, 2009). In accordance of case and

surgical techniques, rehabilitation period, postoperative care and aesthetic look, it is considered better without surgical revision. The aim of this research is to show two surgical cases of dogs with cleft palates, who are implemented in clinic for small animals department of Veterinary medicine in Novi Sad. The valid results will show us post operative care and which technique is better for use.

MATERIAL AND METHODS

In this research we have shown two clinical cases of cleft palates. First case refers to female bull terrier who was at the moment of the procedure 8 months old. The second case refers to male french bulldog who was at the moment of the procedure 4 months old. The surgical intervention in these two patients was considered of good veterinary practical skills, which requires detailed informations from owners and also signing the anesthetic protocol for surgical permission. The preoperative preparation has proceeded basic clinical examination of patients, while we use special diagnostic procedures as using of X rays (Zoomax, EU) and blood samples (biochemical analyses of serum). Both of the dogs are operated in general anesthesia with endotracheal intubation. For premedication we used medetomidin 5µg/kg (Dormitor®, Orion, Hungary), midazolam 0,25 mg/kg (Dormicum®, Roche, Swiss). In purpose of anesthesia induction and maintenance we use propofol 2 mg/kg (Lipuro 1%, Braun, Germany) and sevoflurane (Sevorane, Abot, USA). During the procedure we used 3/0 resorbed suture material (monosorb, Yavo, Pl). The surgical procedure itself implies two different techniques. In first case, the first implied technique was double flapped technique. After 20 days, we did revision with other technique, rotated flap, which has proved better and much stronger. In the case of french bulldog, the rotated flap technique was first implied. The postoperative care was considered by applying antibiotics 25 mg/kg (Synulox®, Haup Pharma, Ita), (Pancef® Alkaloid, MK) and also non-steroid anti inflammatory drugs 2 mg/kg (Rycarfa®, Krka, Slo).



Picture 1. Thoracic X ray of french bulldog (original picture).

RESULTS AND DISCUSSION

In case of bull terrier, first technique which was applied was double flap technique. This procedure started with incision 5 mm from alveolar process and it was parallel with medial suture. The incision was situated across the whole length of hard palate from incisive teeth to the caudal side of tonsils. The mucosa from hard palate was prepared by mobilizing and formed with two flaps which are sutured across the cleft. This sutures are well positioned, directly across the cleft and they closed it in proper way. This patient also required pharyngo gastric catheter. The food has entered direct in stomach by passing the mouth. Unfortunately, the 20th day after the surgery we noticed the dehiscence across whole suture part of mucosa. We found out from owners anamnesis that dog has problem again with swallowing food and drinking the water. We accessed to surgical revision again, but this time with other technique of rotated flap. This technique began with incision on the other side (left) of hard palate. It was located in medial palate suture 5 mm from alveolar process. Flap was prepared and come with rotation under the prepared mucosa of right oral side of hard palate. The earl post operative care was very successful. The catheter was removed after 7 days. We provide required control examinations and the results were above average.



Picture 2. Cleft palate before surgery operation (left picture). On the right side is hard palate after surgical method of double flap technique (original picture).



Picture 3. Surgical technique of rotated flap on bull terrier (original picture).

In second case of french bulldog we instantly applied the second technique of double overlapping flap. The incision was started with mucoperiosteal part of hard palate towards the dental arch 1-2 mm from teeth. Periosteal part with mucosa was prepared and his role was to create overlapping flap (Fossum, 2007). The overlapping flap was rotated and underlined under prepared mucosa on the other side where we connect tissue with single knotted suture. This patient also had pharyngeal gastric catheter. After one month, control examination showed us good basic results of wound and sutures.



Picture 4. On the left side, surgical technique of rotated flap. Cleft palate look before surgery right side.

Martinez- Sanz and co. (2011) discuss about non invasive methods in healing this problem. They pointed out from research that mushed food has very big role. Owners should apply every

day food to bottles with proper extensions for weaning. It is also indicated to use orthodontic extensions which are made of silicon and plastic materials and directly cover the cleft. These methods are only for temporary use and they require specific steps of hygiene prevention in oral cavity. Campbell and Nickson (1965) considered also that in palatoplasty with two mucoperiosteal flaps we achieve not only good closing defect, but for safety reasons prevention from dehiscence, faster healing and formation of new bone.

CONCLUSION

Based on the surgical comparison of two techniques, double flap and rotated flap in purpose of healing clefts in two dogs, we can determine:

1. Applying double flap technique cause the suture dehiscence and revision of surgical technique was necessary.
2. Applying overlapping flap technique as first surgical procedure in case of french bulldog, and as second technique in case of bull terrier brought us to aim of research- closing the cleft.
3. The sutures which were directly posted across the cleft, brought the tissue of mucosa to dehiscence.
4. Technique of rotated flap brings better mechanical fixedness for sutured mucosa of hard palate.

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INFLUENCE OF DIFFERENT ANTIOXIDANT AGENTS ON THE AROMATIC PROFILES OF SAUVIGNON WINES

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Summary: During the harvest 2018, vinification of Sauvignon Blanc grapes (*Vitis vinifera* L.) was carried out with 4 different enological antioxidant preparations, containing lower concentrations of SO₂ in combination with ascorbic acid and different enological tannins in comparison to 2 treatments with sulfuric acid, in the must prior to alcohol fermentation. The basic parameters of quality, free amino-nitrogen (FAN) and total polyphenolic compounds were spectrophotometrically analysed in the must. Basic physicochemical parameters and individual volatile aromatic compounds in wines were determined using the gas chromatographic method. The obtained results undoubtedly pointed out the positive effect of sulfuric acid on the dynamics and the outcome of alcohol fermentation with the lower dose influencing the highest concentration of total aromatic compounds. Due to the use of combined antioxidants, alcohol fermentation did not totally complete, while the combination of potassium pyrosulphite, L-ascorbic acid, gallotannins and ellagitannins, resulted in the highest concentrations of aromatic monoterpenes and fruity esters in wine.

Key words: antioxidants, aromatic compounds, gas chromatography, Sauvignon Blanc, wine

INTRODUCTION

Sulphur dioxide (SO₂) is the most widespread preservative in the world and commonly used in the wine production due to its antioxidative, antiseptic and coagulation properties. While moderate oxidation improves the quality and sensory characteristics of red wines, SO₂ is essential for the preservation of the color and aromas, as well as microbiological stabilization of the white wine. Due to its positive conservation and regeneration effect on the wine aromas, SO₂ is an essential addition of wine production (Chinnici et al., 2013). The main antioxidative function of SO₂ is binding to hydrogen peroxide, which is the product of O₂ reduction. In that process, it inhibits the aldehyde production and prevents the oxidation of other readily oxidizable compounds (Boulton et al., 1996; Elias and Waterhouse, 2010). Moreover, SO₂ also reduces quinones (brown polymers), the results of phenol oxidation, and in the same time improves polyphenolic wine composition (Danilewicz et al., 2008).

Despite its positive properties, SO₂ may often cause allergic reaction in certain segments of population (European Commission, 2003). SO₂ may induce the increase the appearance of chromosome abnormalities as well as sister chromatid and micronuclei exchange in the mammalian cells. Also, researches shown that it causes oxidative damages on the both male and

female organs of the mouse (Meng et al., 2004)., which is the reason why International Organization of Vine and Wine gradually decreases maximum permitted levels of SO₂ in wine (OIV, 2012). The total permitted quantity of SO₂ in red wine is up to 150 mL/L, and up to 200 mL/L for the white wine, (European Commission, 2009). Contemporary trends that include healthy lifestyle mean that consumers are looking for healthier and higher quality products. Due to increase of demands for preservatives-free products, winemakers are facing one of the biggest challenges of modern wine-making- how to find healthier alternatives that could replace antimicrobial and antioxidant properties of sulphites, and in the same time to protect the sensory characteristics of the wine.

There are many studies about antimicrobial replacement of SO₂; however, antioxidant properties are still inadequately researched. Moreover, ascorbic acid has been lately seen as an adequate alternative for the SO₂ since it is more active with molecular oxygen, but it affects the color development in white wines, which means that one should be careful with its usage (Bradshaw et al., 2011). In order to improve chemical and sensory characteristics of the white wine, oenological tannins are commonly added. Hydrolysable tannins, such as gallotannins and ellagitannins are extracted from oak and have significant antioxidant capacity. On the other hand, condensed tannins are obtained from grapes (Haslam, 2007).

Wine aroma is a rather complex feature, formed by aromatic compounds from grape as well as of compounds formed during and after the alcoholic fermentation. So far, a couple of hundred aromatic components have been identified. They have a wide range of concentrations that are commonly expressed in mgL⁻¹, µg/L or ng/L (Rapp, 1988). Ribéreau-Gayon et al. (2006) developed a classification of white wines, which includes three different varieties: neutral white wines, Chardonnay varieties, Sauvignon varieties and aromatic wines. They also claim that aromatic cultivars are wine varieties whose scent is formed by the primary aroma compounds. Most often, those are terpenes, with the exception of Sauvignon. Due to its lower concentration in grapes and wine, compounds such as methoxypyrazine and volatile thiols, typical for Sauvignon, have been discovered relatively late (Bayonove et al., 1975; Darriet et al., 1993) and their complex analysis are still relatively rarely done. Terpenes are plant metabolites with specific flowery scent. Grapes (*Vitis vinifera* L.), contain monoterpenes, a subgroup of terpenes, that are responsible for flowery aromas of muscatel cultivars as well as for the scents of aromatic cultivars eg. Traminer. The most important representatives are linalool, nerol, geraniol, citronellol and α-terpineol (Câmara et al., 2007; Jackson, 2008). Aromas of the terpenes are mostly characterized as floral- they include the scent of rose, coriander, camphor, green and herbal scent (Simpson, 1979; Güth, 1997). Higher alcohols appear due to yeast metabolism together with ethanol and therefore form up to 50% of the wine's aromas related with the scent of ripe fruit, flowers and roses (Jackson, 2008; Palomo et al., 2006). The vast majority of secondary alcohols have got sharp smell similar to the smell of chemicals and solvents. Their total concentration in white wines is rarely more than 300 mgL⁻¹, which is considered to be an upper limit for the positive aroma development (Usseglio Tomasset, 1975; Jackson, 2008). Esters are important aromatic compounds of wines, while fruit characteristics of certain esters are

significant for the development of young wine's aroma. The majority of aromas are developed by the yeast during the process of alcohol fermentation.

There are several studies about the influence of alternative preparations compared to sulphites on the antiseptic characteristics and the aroma of white wines (Chinnici et al., 2013; Ferrer-Gallego et al, 2017). However, we still do not have enough data to know in what measure does the ascorbic acid or any other oxidant alternative influence the color and aroma of the wine. Therefore, this paper deals with the usage of different antioxidants with lower concentration of SO₂ together with compounds such as ascorbic acid and different types of tannins. Physicochemical and instrumental analysis have been used to monitor the influence of different agents on the essential composition and concentration of aromatic compounds in Sauvignon wines compared to the usage of different concentrations of sulphites, before the process of alcohol fermentation.

MATERIAL AND METHODS

Grape variety involved in this experiment was Sauvignon Blanc, cultivated and processed at the Jazbina Experiment Station, Faculty of Agriculture, Zagreb. Sauvignon is a white wine variety used for aromatic and dry wine production. With its pronounced and refreshing aroma, it is one of the most important wine varieties in the majority of the world's wine regions. It has got distinctive green paper, asparagus, elderflowers and boxwood aroma. However, its aroma is influenced by the climate conditions- while wines from warmer, tropical areas have tropical, fruity aroma, wines from the colder areas have herbal, earthiness aroma. Higher quality clones often have subtle floral aroma (Jackson, 2008). Sauvignon wines were produced by the processing of 150 kg of technologically mature grapes. Manual grape harvesting was conducted on the September 6th, 2018. The grapes were processed by using automatic destemmer and crusher. Pressing was done by using of hydraulic press. After the pressing, the researchers determined sugar concentration, total acidity, pH, free α -amino nitrogen and total phenols in the must. The must was divided in six different inox 20 L-vassels (sample A, B, C, D, E, F). Afterwards, different oenological antioxidant agents have been added to each sample (Table 1). After 24-hour sedimentation, clear must have been separated from the sediment using closed pumping (with plastic hose) and put into 10 L-glass tanks (in duplicate).

Table 1. Treatments in production of Sauvignon 2018.

Treatments	Antioxidant	Dose
A	Sumpovin (H ₂ SO ₃), Inovet d.o.o.- control	20 ml/20 L
B	Sumpovin (H ₂ SO ₃), Inovet d.o.o.	10 ml/20 L
C	Aromax® (K ₂ S ₂ O ₅ + ascorbic acid), AEB	4 g/20 L
D	Aromax Super® (K ₂ S ₂ O ₅ + ascorbic acid + proanthocyanidins + ellagitannins), AEB	4 g/20 L
E	Aromax Gal® (K ₂ S ₂ O ₅ + ascorbic acid + gallotannins), AEB	4 g/ 20 L
F	Noxitan® (K ₂ S ₂ O ₅ + tannins), AEB	2 g/20 L

In the experiment, five different oenological agents were used. The purpose of these agents is to preserve white wines aromas and prevent oxidation in the early stages of the processing, i.e., before the fermentation. After the separation of the must, the next step was the inoculation with selected yeast (2 g/10 L) Excellence TXL® (Lamothe Abiet) and adding the activator of the yeast (3 g/10 L) Oenostim® (Lamothe Abiet). Complex yeast nutrition Vitaferment® (Lamothe Abiet) based on ammonium, salt and vitamin B1 was added to help the yeast reproduction and decrease anomalies in the fermentation's dynamics conducted at 15°C.

Sumpovin is a 5-6% sulphuric acid that can be used, depending on the dosage, for accelerating the sedimentation in must (soil particles, moulds, pesticides etc.). It postpones the process of fermentation by blocking the development of microorganisms such as bacteria and wild yeasts. Together with antioxidative properties, i.e., keeping grape juice and wine from getting brown, it improves the wine quality (<http://www.sumpovin.com/sumpovin>). In this experiment, it was used as a control (A) in the regular dosage of 100 mL/100 L and as treatment B in the half lower dosage. Aromax® contains 50% of potassium pyrosulphite that releases 54% of SO₂ and 50% of L-ascorbic acid. It protects must and primary aromas (terpenes) from the oxidation during the process of primary grapes processing by removing the oxygen from the must. Extensive researches have shown that must with up to 8 ppm of oxygen may reach values close to zero in only 36 hours after the primary processing. Sauvignon wines yield particularly good results-samples processed by using Aromax have shown much more intensive aromas of green pepper, asparagus, figs and grapes (<http://www.cedar-agro.hr>). Aromax Super® is consisted from 50% potassium pyrosulphite, 35% L-ascorbic acid and 15% pure gallotannins. These elements enable the control of peroxides and acetaldehyde that are developed by oxidative enzyme activity and therefore protect white wine aromas. This product decreases the concentration of SO₂ and in the same time increases the dosage of free SO₂ (<http://www.ireks-aroma.hr>). Aromax Gal® represents 50% of potassium pyrosulphite, 35% of L-ascorbic acid and 15% of pure gallotannins. It protects white musts from changing their colour and inhibits activity of polyphenol-oxidase such as laccase and tyrosinase. It binds the oxygen in a short period of time and inhibits the development of spontaneous fermentation (<http://www.ireks-aroma.hr>). Noxitan® is a

combination of potassium metabisulphite and tannins with higher antiradical activity. It is also extremely efficient in preventing oxidation. The ellagitannins used increase antiseptic activity of SO₂ and reduce its use for about 20%. Noxitan prevents the development of peroxids and protects the colour and primary aromas. It increases the content of free and molecular SO₂, which as a result, improves antiseptic activity and preserves sensory characteristics (<http://www.ireks-aroma.hr>).

Laboratory analyses were conducted in November 2018 in the Laboratory for Grape, Must and Wine, Department of Viticulture and Enology, Faculty of Agriculture, University of Zagreb, after the process of alcoholic fermentation. The researchers analyzed the specific weight, alcohol percentage, total sugar amount, total and volatile acids, extracts, ash, SO₂ and pH of the given samples (method prescribed by O.I.V., 2012). Free α -amino nitrogen (FEN) was determined using the spectrophotometer „Specord 400“, Analytik Jena, Jena, Germany, using the method proposed by Dukes and Butzke (1998). Total phenolic compounds in must and wine were spectrophotometrically determined using the Singleton and Rossi method (1965), based on the color reaction of the phenolic compounds with the Folin-Ciocalteu reagent. There were three measurements in total, while the mean value has been finally taken. The results were expressed in gallic acid equivalents (mg GAE/L). Volatile compound extraction in the wines has been conducted by using the solid phase extraction with LiChrolut EN columns (500 mg, 3 mL, Darmstadt, Germany). Before the sample application, column was conditioned with 3 ml of dichloromethane, methanol and 13% ethanol aqueous solution. At the end of the conditioning process, 50 ml of wine sample was added. After the extraction, it was left to air dry for 30 minutes. Then the analytes were rinsed with 1 ml of dichloromethane. Used column was 60 m x 0,25 mm i.d. x 0,50 mm ZB-WAX cartridge (Phenomenex, SAD). The analysis was conducted on the GC-MS system (gas chromatographs Agilent Technology 6890, mass spectrometer Agilent 5973N).

Mean values of concentrations and their standard deviations were calculated from two replicates. One-way analysis of variance (ANOVA) was performed using the SAS System for Windows 9.0, 2004 (SAS Institute Inc., USA). The differences in the content levels were estimated with the t-test. Statistically significant were considered $p < 0.05$. The results of Principal components analysis (PCA) presented as two-dimensional PCA plots were used to identify the differences between wines.

RESULTS

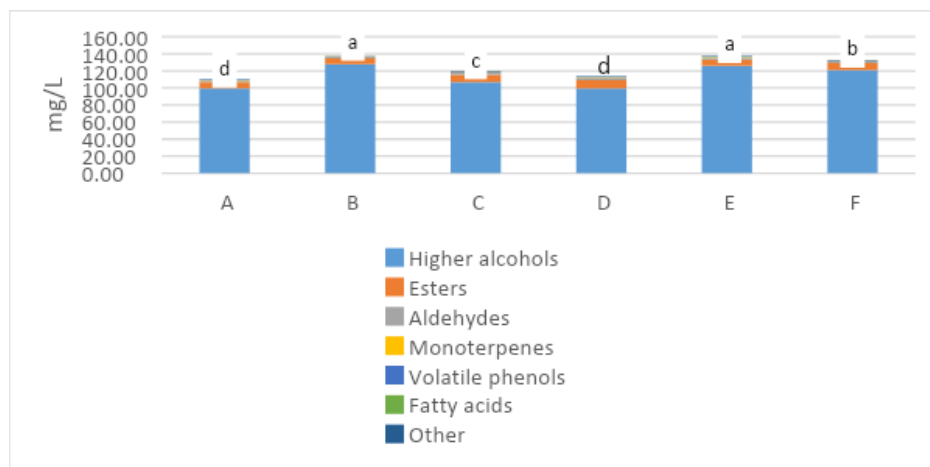
After the primary processing and deposition, the sugar concentration in the must samples of Sauvignon was determined and results are shown in the Table 2, together with concentration of free α -amino nitrogen (FAN), pH value and the initial concentration of the total phenols. The results of basic physicochemical analysis are shown in the Table 3. Besides the basic parameters, the researcher also conducted the analysis of the individual volatile aromatic compounds of the wine (data not shown), divided into seven different groups (Graph 1).

Table 2. Sugar concentration, pH, total acidity, free α -amino nitrogen (FAN) and total phenols in must

Sugar (°Oe)	pH	Total acidity g/L (as tartaric)	FAN γ (AN)/mgL-1	Total phenols γ (UF)/mgL-1
95	3,34	7,4	28,49	340,35

Table 3. Basic physicochemical parameters of wine Sauvignon 2018.

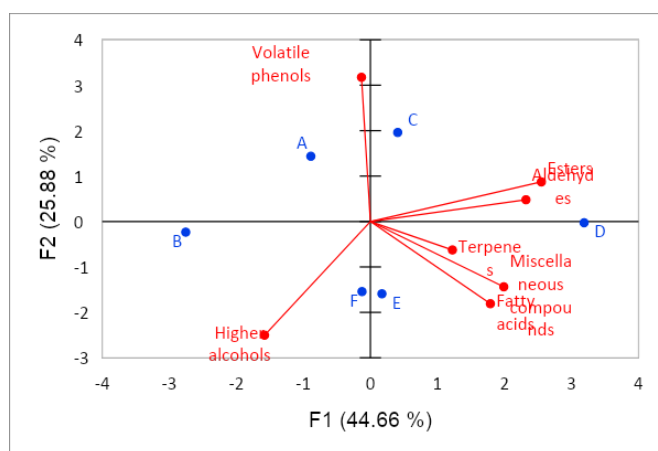
Parameter	A	B	C	D	E	F
Specific gravity (20/20°C)	0,98993	0,9903	0,9914	0,9914	0,9908	0,9912
Alcohol (gL ⁻¹)	108,5	107,8	107,1	106,4	108,5	107,8
Alcohol (vol%)	13,7	13,6	13,6	13,5	13,7	13,6
Total extract (gL ⁻¹)	19,8	20,6	23,2	22,9	22,2	22,9
Residual sugars (gL ⁻¹)	2,8	2,7	5,8	6,2	5,3	5,6
Sugar-free extract (gL ⁻¹)	18,0	18,9	18,4	17,7	17,9	18,3
Residual extract (gL ⁻¹)	12,9	13,0	13,2	12,7	12,5	12,6
Total acids (as tartaric) (gL ⁻¹)	5,8	6,6	6,0	5,9	6,2	6,4
Volatile acids (as acetic) (gL ⁻¹)	0,60	0,54	0,65	0,70	0,65	0,59
Non-volatile acids (gL ⁻¹)	5,1	5,9	5,2	5,0	5,4	5,7
pH	3,33	3,28	3,30	3,30	3,28	3,31
SO ₂ free (mgL ⁻¹)	0	0	0	0	0	0
SO ₂ bound (mgL ⁻¹)	37,0	40,0	59,0	55,0	57,0	45,0
SO ₂ total (mgL ⁻¹)	37,0	40,0	59,0	55,0	57,0	45,0
Ash (gL ⁻¹)	1,71	1,74	1,73	1,72	1,72	1,78



Graph 1. Proportion of aromatic compound groups in the aromatic profile of Sauvignon 2018. Total concentrations labeled with different letters (a-d) are significantly different at $p=0.05$.

DISCUSSION

The results of the must analysis are shown in the Table 2. As a result of a very warm wine-growing year (2018), regardless of the early harvesting, relatively high sugar concentration was found in the must. It indicates a higher alcohol percentage- > 13 vol % in the final product. Sauvignon must contained optimal amount of total acidity, while pH value was within the range that is optimal for the white wines (from 3.1 to 3.4) (Jackson, 2014). Those results were expected due to the specific wine variety and the climate, and it is also an indicator of freshness and microbiological stability. The free α -amino nitrogen (FAN) concentration points to the undernourishment of the must and reminds us of using the yeast nutrition, which provides the regular course of the alcoholic fermentation and clean aromas of the wine.



Graph 2. Two-dimensional PCA plot of the first and second principal components after the PC analysis of aromatic compound groups in Sauvignon wines 2018.

The concentration of total phenols (TP) was relatively high compared to some other concentrations found in different musts of some varieties in Croatia. According to Jagatić Korenika et al. (2015), the range of the TP in white wines was between 92 and 383 mgL⁻¹. According to Jakobović et al. (2012) the concentration of TP in Riesling from Croatia ranged between 394 and 482 mgL⁻¹.

Analyses of all wine treatments are shown in Table 3. Taking sugar concentration of the must into account, all the wines contained alcohol concentration in the range between 13.5 and 13.7 vol %. Only samples A and B were dry types of wine (< 4 g/L of sugar), while all the other treatments gave semi-dry wines (>4 g/L of sugar). Due to the fact that the same type of base wine and the same yeast were used, the difference in the fermentation's dynamics can be prescribed to the fact that different antioxidant preparations were used. Sugar-free extract was in the range between 17.7-18.9 gL⁻¹, while the highest value was registered in the sample B. Extract is a very important component of the wine's quality, which hugely improves its fullness and harmony and is usually in a range between 15 and 39 gL⁻¹. However, its range for the dry white wines is usually below 25 gL⁻¹ (Ribéreau-Gayon et al., 2000). The total acidity (TA) was between 5.8 - 6.6 gL⁻¹. Because of the tartrate sedimentation during the alcoholic fermentation, it is expected to have lower TA compared to the initial must concentration. The highest concentration was measured in the sample B, while the lowest one in the A variant. One should take into consideration that both samples were treated using the sulphuric acid in different dosages. The range of volatile acid's concentration is between 0.54 and 0.70 gL⁻¹, which is according to the values prescribed in the Regulations of the Wine Production (N.N. 2/2005). The ash in the wine is formed by the minerals from the solid parts of the grapes. The concentration of the ash in the wine ranges from 1.5 to 3 gL⁻¹ (Ribéreau-Gayon et al., 2000). Due to the fact that white wines are mostly produced without the maceration, they usually contain only minimal amounts of the ash. This research showed the results between 1.71 and 1.78 gL⁻¹. Considering the fact that the usage of sulphites in different dosages and combinations is the main aim of this work, it is important to note that the results of the free, bound and total SO₂ in the wines after the fermentation show the expected results- the absence of the free SO₂. Bound SO₂ ranged from 37-59 mgL⁻¹, while the concentration increased in following way: A>B>F>C>D>E. The total SO₂ amount meets the regulations prescribed in the Regulations of the Wine Production (N.N. 2/2005), while in the same time, more SO₂ should be added due to the lack of the free SO₂.

In total, 92 volatile compounds were analysed. They were divided into 6 chemical groups and one that includes miscellaneous compounds. The highest number of compounds were analysed in the group of higher alcohols (24), then esters (21), monoterpenes (17), volatile phenols (8), aldehydes (5), fatty acids (5) and the other compounds (9). The concentration of the total aromatic compounds increased following the succession: A>D>C>F>E>B (Graph 1). The highest number of individually significant differences among volatile compounds was registered in the treatment D (Graph 2). The most represented group of the individual compounds were the higher alcohols, which is in accordance with previous research (Jagatić Korenika et al., 2015). The highest concentration was in the sample B. In this research, the total concentration of the higher

alcohols were in the range 99.45-127.94 mgL⁻¹, which means that it still has a positive impact on the wine's aromas. The most representative higher alcohols with concentrations above the sensory perception threshold were phenyl ethyl ethanol (the highest concentration in samples B and E), 2-methyl-1-butanol (with no statistical difference among all samples), 1-hexanol (the highest concentration in sample B). The scent of isoamyl alcohol is related to whiskey, solvent or nail polish, phenyl ethanol reminds on flowers and rose, 2-methyl-1-butanol on black truffle and 1-hexanol on freshly cut grass (Jackson, 2008). The highest concentration of fruity esters significant for the white wine's aromas, is found in sample D, while the lowest in sample A. The most representative ester in all variants was ethyl-hydrogen succinate which can influence the wine's acidity due to the higher concentration of succinic acid in wines. Esters of the succinic acid are used to imitate the smell of the butter, rum, brandy, grapes and raspberries (Patel and Shibamoto, 2003). The second most representative ester was the isoamyl acetate with banana scent, then diethyl succinate with wine and apples aroma, ethyl octanoate with the scent of pears and ethyl decanoate with sweet-fruity aroma. According to Jagatić Korenika et al. (2015) the most representative ester in white wines from Croatia, after the ethyl acetate, was the isoamyl acetate with the concentration between 0.31-4.14 mgL⁻¹, then the ethyl hexanoate and ethyl octanoate.

The group of monoterpenes, as representatives of the variety's aroma, were equally present in all samples. The most common compound was nerol, with the highest concentration in the samples B and D, and geraniol in the sample D. The nerol's scent is described as flower/citrus, while the scent of geraniol as flowery (rose). This research showed different results compared to the results of the indigenous white varieties in Croatia, where the most representative monoterpene was linalool with citrus-flowery aroma (Jagatić Korenika et al., 2015). β -damascenone, a representative of C13-norisoprenoid, with its flowery aroma significantly improves the aroma of Sauvignon, was also analyzed. The concentrations were above the threshold, while its highest concentration was in the sample F. The most abundant compound in the group of volatile phenols was tyrosol (11-23 mgL⁻¹), a natural antioxidant present in wines, as the product of the yeast's metabolism. It is also present in the olive oils and has a positive impact on human health. The highest concentration was in the sample A. If it appears in still wines above 25 mgL⁻¹, it can cause a bitter taste (Jackson, 2009). However, it is more common in sparkling wines, after the secondary fermentation. The second most represented compound was the compound with a pleasant jasmine and almond taste- it is a benzyl alcohol with the highest concentrations in the samples A and B. The most significant fatty acid present in this research was propionic acid with the highest concentration in the sample D. All the fatty acids analyzed in this research were present in the concentrations below the sensory perception threshold that usually has a positive impact on aroma complexity. The highest concentration of the total aldehydes was in sample D, while the most abundant compound was 2-octenal with its nutty aroma.

The marked differences in the concentrations found between the individual samples were confirmed by the PCA analysis, where the clustering of individual samples was evident according to the aromatic compound groups (Graph 2).

CONCLUSION

Based on a one-year experiment with Sauvignon Blanc wines treated with different oenological antioxidants and analysis of must and wine, following conclusions can be made. Only samples treated with sulphuric acid resulted in dry wines and the lowest concentration of the bound SO₂ after the alcoholic fermentation. There was statistically significant difference in the total aromatic compounds concentration among the samples. The highest concentration was registered in the sample B and E the lowest in the sample A, so it can be concluded that a different dosage of sulfuric acid had a significant effect on the aromatic profile of Sauvignon wine. The most abundant group of aroma compounds in all samples were higher alcohols with isoamyl alcohol as the most important compound with significantly highest concentration in the treatment B. There was no significant difference in the concentration of total monoterpenes. The most common terpene with its floral-citrus aroma was nerol with the highest concentration in samples B and D. β -damascenone, as representative of C13 norisoprenoids, differed significantly with the highest concentration in the sample F. Significant difference was observed in the fermentation aromas, such as total esters. Sample D (Aromax Super®) contained the highest number of individually represented fruity esters. Due to the use of combined antioxidants, the alcoholic fermentation was not completely finished. Combination of potassium pyrosulfite, L-ascorbic acid, gallotannins and ellagitannins resulted in the highest concentrations of monoterpenes and fruity esters. Following research on the use of alternative antioxidants in wine should, as addition to the existing analyses, include sensory analysis of wine.

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SENSITIVITY OF *CERCOSPORA BETICOLA* SACC. ISOLATES TO AZOXYSTROBIN

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Summary: Testing the sensitivity of *Cercospora beticola* Sacc. isolates to azoxystrobin, by selecting different isolates from sugar beet and a control isolate from beet, representing the wild population of *C. beticola* and by applying different concentrations of azoxystrobin in combination with SHAM to the isolates. The purpose of this study was to prove that *C. beticola* is resistant to azoxystrobin due to inadequate and long term fungicide applications and finding new ways to protect affected plants from this pathogen.

Key words: sugar beet, azoxystrobin, *Cercospora beticola*, resistance.

INTRODUCTION

Cercospora beticola Sacc. is a pathogenic fungus that causes leaf spot disease in sugar beet and it is the most economically significant disease of sugar beet leaves. Hosts are plants of the genus Beta (sugar beet, beet, chard), Spinacia (spinach), Lactuca (lettuce), etc. It originates from Central Europe and the Mediterranean, but is widespread in all sugar beet growing regions (Ivanović i Ivanović, 2001).

Cercospora beticola Sacc. taxonomically belongs to Fungi kingdom, Deuteromycetes subdivision, Hyphomycetes class, Hyphales order, *Cercospora* genus (Ivanović i Ivanović, 2001).

In its life cycle, *C. beticola* overwinters in form of stromatas and conidias in infected plant remains. In favorable humidity and temperature conditions, conidiophores and conidias are formed on stromas and they are spreaded by wind and rain drops (Budakov i sar., 2014). Symptoms appear first in the older and middle leaves of the rosette and then on the younger ones. Round spots of greyish color, surrounded with a dark red ring which is characteristic for this disease. With the development of the disease, the number of spots increases, leaves dry and die off and retrovegetation occurs. In the necrotic part of the spot appears a gray coat of conidiophores and conidias. The tissue within the spots necrotizes and falls out, so the hollowness of the leaves occurs as a secondary symptom. *C. beticola* beside leaves infects leaf petiole, inflorescence, flower parts, trunk and fruit (Balaž i sar., 2010).

Azoxystrobin belongs to the group of strobilurins (QoI fungicides) which mechanism affects the inhibition of mitochondrial respiration by bonding to ubiquinol oxidase (Qo) on cytochrome b (FRAC, 2018). This reaction inhibits spore and mycelium growth.

Fungi in the presence of strobilurin in the substrate have the ability to use alternative oxidase (AOX), which allows sensitive isolates to achieve mycelium growth on substrates with added fungicide (Avila-Adame i Köller, 2003). In this case resistance occurs.

When it comes to in vitro studies on the sensitivity of *C. beticola* isolates to strobilurins, methods based on measuring the inhibition of spore germination (Secor i sar., 2010) or determining the reduction of mycelial growth are applied (Malandrakis i sar., 2011). In this study, salicylhydroxamic acid (SHAM) was added to the substrate. It inhibits AOX activity during the strobilurin resistance test, affecting mycelial growth of sensitive isolates, revealing only the resistance caused by a mutation in cytochrome b (Larson, 2004).

AIM OF RESEARCH

Determination of the amount of active substance required to inhibit mycelial growth in vitro, collecting isolates of *Cercospora beticola* Sacc. and testing the sensitivity to the active substance azoxystrobin.

MATERIAL AND METHODS

The location of characteristic symptoms was a field of sugar beet in Rimski Šančevi. Leaves were collected and brought to the Phytopathology Laboratory at the Department of Phytomedicine and Environmental Protection of the Faculty of Agriculture in Novi Sad.

A. Isolation of the pathogen and forming a collection of isolates

To achieve monosporial isolates, individual conidias were obtained from spots on sugar beet leaves, which had previously been washed under running water and incubated in a humid environment for 24-36 hours at a temperature of 25 ° C. Conidias were transferred to the nutrient medium (PDA) with an entomological needle. Incubation of the colonies was performed in an incubator at 25 ° C for 4 days. The formed mycelium was screened on a new (PDA) medium and grown for 7 days under same conditions. To form a collection, 10 isolates (with 3 repetitions) were obtained, with one beet isolate from organic production used to represent the wild population of *C. beticola*. The isolates were assigned codes in a combination of letters and numbers (Table 1).

Table 1: List of the isolates used in the examination

Serial number	Code of the isolate
1	2B
2	4A
3	5A
4	6A
5	6B
6	7A
7	8A
8	8C
9	9
10	11A
11 (beet)	CB cvekla A

B. Testing of *C. beticola* isolates sensitivity to azoxystrobin

Different concentrations of azoxystrobin supplemented with salicylic hydroxamic acid (SHAM) were used (Table 2.). Technically pure azoxystrobin (96%) was dissolved in methanol, and after cooling the substrate, a certain amount of this solution was added to achieve the indicated concentrations of azoxystrobin in the substrate. Mycelial fragments (3) with 5 mm diameter were placed on the supstrate with added fungicide in three repetitions.

Table 2: Concentrations of azoxystrobin supplemented with SHAM

Concentration of azoxystrobin in supstrate	Concentration of SHAM
0,01µg/ml	0,5mM
0,1 µg/ml	0,5mM
0,5µg/ml	0,5mM
1µg/ml	0,5mM
Control	0,5mM

Petri dishes with mycelia fragments were incubated at 25 ° C for seven days. The results were read by measuring two normal colony diameters. For each isolate and test concentration was calculated the relative mycelium growth, based on the value of colony diameters in Petri dishes with added fungicide and without added fungicide (control) by formula:

$$RG = \frac{RF}{RC} \cdot 100 (\%)$$

RG - Relative mycelium growth

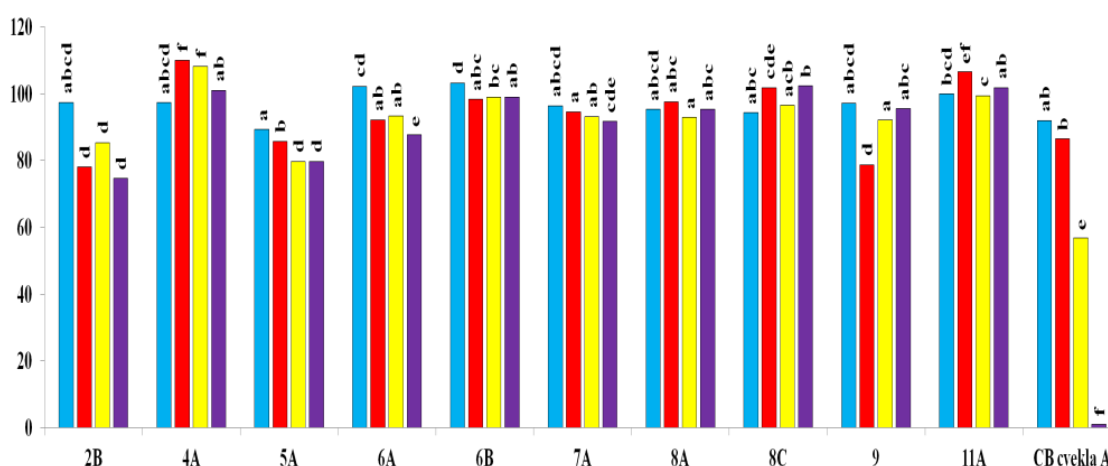
RF - Average diameter of the mycelium on the substrate with the fungicide (mm)

RC - Average diameter of mycelium on control - no fungicides (mm)

Statistic was done using analysis of variance (ANOVA), and the significance of difference between pairs was done using Fisher LSD (least significant difference) test.

RESULTS AND DISCUSSION

The results of testing the sensitivity of *Cercospora beticola* isolates to azoxystrobin are presented in the chart.



Concentration of azoxystrobin in substrate:

First column: 0,01 µg/ml

Second column: 0,1 µg/ml

Third column: 0,5 µg/ml

Fourth column: 1 µg/ml

At the lowest observed concentration the statistical differences are generally non-existent, except in some cases, for example where isolate 5A had a significantly lower relative value than isolate 6A. As the azoxystrobin concentrations increase, the differences in relative values between isolates increase too, so that at concentrations of 0,1 µg/ml and 0,5 µg/ml the isolates can be grouped into those with the lowest relative increase (2B) to those with the highest (4A). At the highest concentration, in all isolates the tendencies in change of relative increase remained the same as in previous concentrations (0,1 µg/ml and 0,5 µg/ml), while only in beet from organic production the relative increase was almost barely registered and was significantly lower compared to other isolates.

Resistance to fungicides is first observed when the expected level of protection is not achieved by the recommended dose. The risk of developing resistance is high, because *C. beticola* has a high potential of conia production, usually has several development cycles and therefore a high part of the population is exposed to fungicides, which is crucial for rapid developing of resistance. In order to solve the problem of resistance, it is suggested to use combinations of systemic and protective fungicides with different mechanisms of action, to use them at recommended doses, applying integral plant protection and other measures of anti-resistance strategy. If resistance occurs, it will be necessary to introduce new active substances with different mechanisms of action from the existing ones, which is considered to be difficult due to high costs of research and implementation, as well as lack of new molecular structures that could become pesticides with new mechanism of action.

CONCLUSION

The results of determination for the sensitivity of *Cercospora beticola* Sacc. isolates to azoxystrobin indicate a high percentage of resistance to this fungicide, compared to the control. All of the *C. beticola* isolates from sugar beet had shown high resistance on different concentrations of azoxystrobin (0,01µg/ml; 0,1 µg/ml; 0,5µg/ml; 1µg/ml). Only the isolate from organic beet (used to represent the wild population of *C. beticola*) had shown moderate sensitivity to azoxystrobin in concentration of 0,5µg/ml and high sensitivity to azoxystrobin in concentration of 1µg/ml. The risk of developing resistance is high, which is why combinations of systemic and protective fungicides must be applied.

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